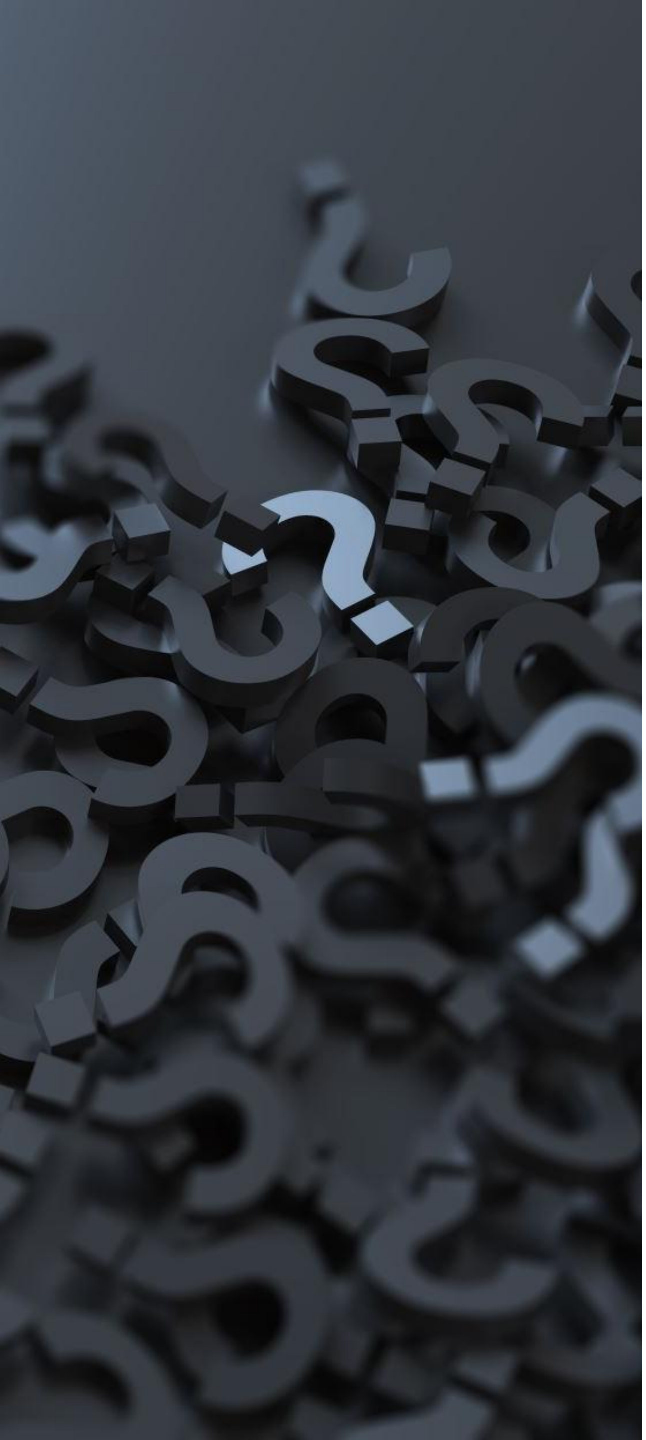
Data Warehouses

Lecture 7



Reminder

Last meeting:

- Multidimensional model - logical model – facts, measures and dimensions
- Data cube and it’s representations
- OLAP schemas

Lecture Goals

Goal:

- Multidimensional model
  - *Drilling into data using dimensions*
  - *Measures and aggregates*
- OLAP operations
  - *What can we do with a data cube?*
- OLAP architectures

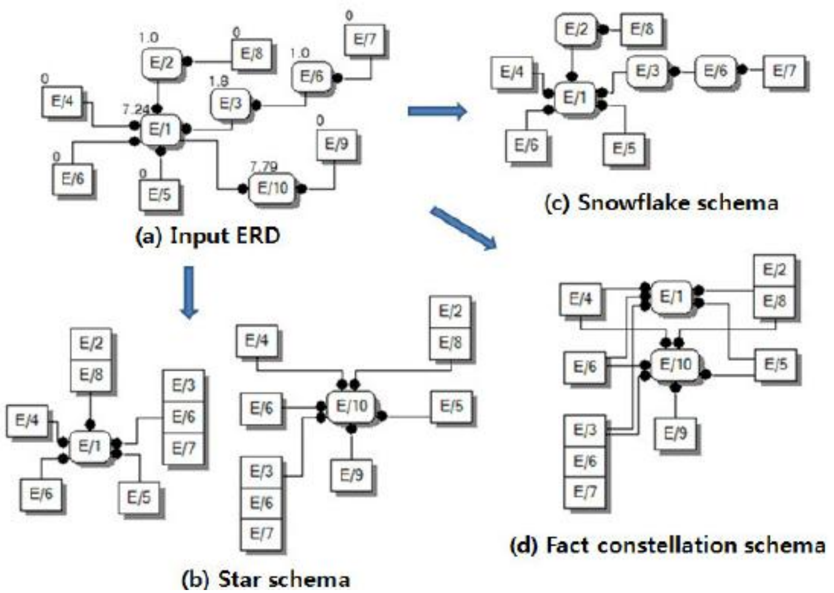
Dimensional and ER

ER Model

- OLTP systems
- Data consistency, non-redundancy, and efficient data storage are critical
- Entity-Relationship Modelling
  - *Removes data redundancy*
  - *Ensures data consistency*
  - *Expresses microscopic relationships*

Dimensional Model

- Focus is on how managers view the business
- Information is centered around a business process – how the business measures the process in many ways along several business dimensions
- Dimensional Modelling
  - *Captures critical measures*
  - *Views along dimensions*
  - *Intuitive to business users*



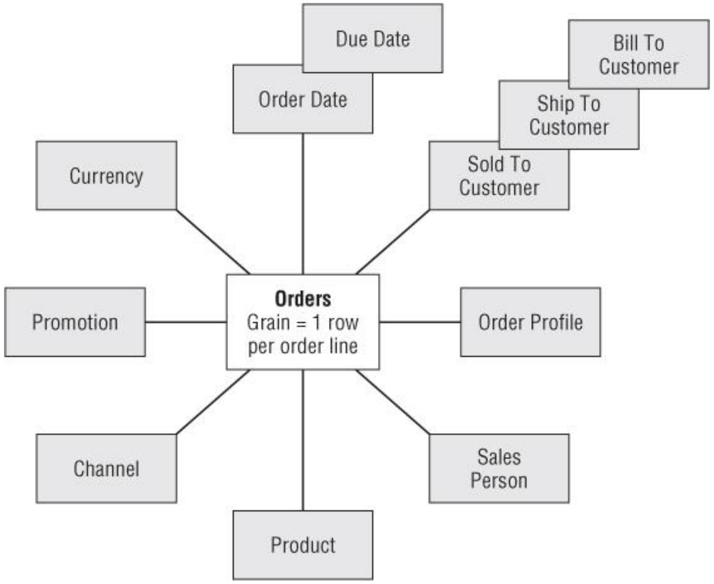
Facts and Dimensions

Fact:

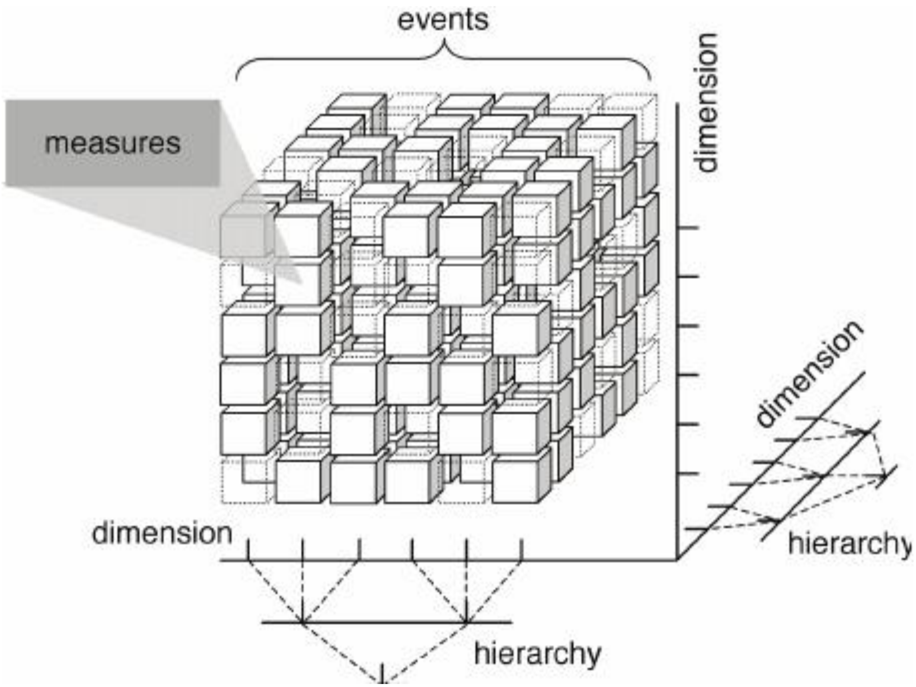
- A fact is something that has happened or been measured.
  - *Like sale of a single product to one customer or the total amount of sales of a particular item in a month.*
  - *fact is a numerical value*
  - *usually associated with several dimensions*

Dimension:

- A dimension is the main analytical object in a BI space.
  - *can be a list of products or customers, time, geographic location, or any other entity that is used to analyze numeric data.*
- Dimensions are related to facts.
  - *The reason for their existence is the addition of qualitative information to the numerical information contained in the facts.*
  - *Sometimes a dimension may have to do with other dimensions, but directly or indirectly it will always be somehow related to the facts.*

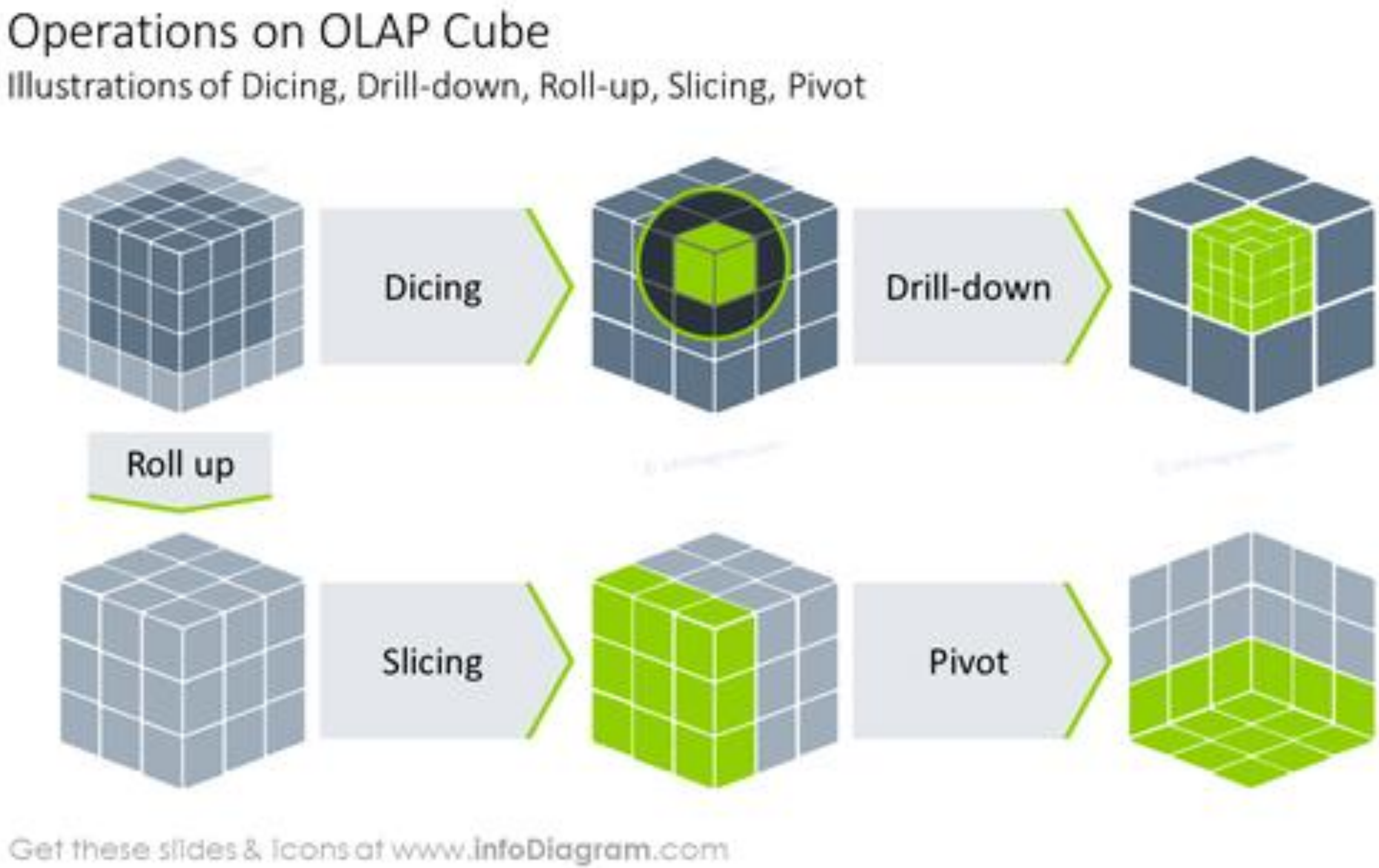


Multidimensional Data

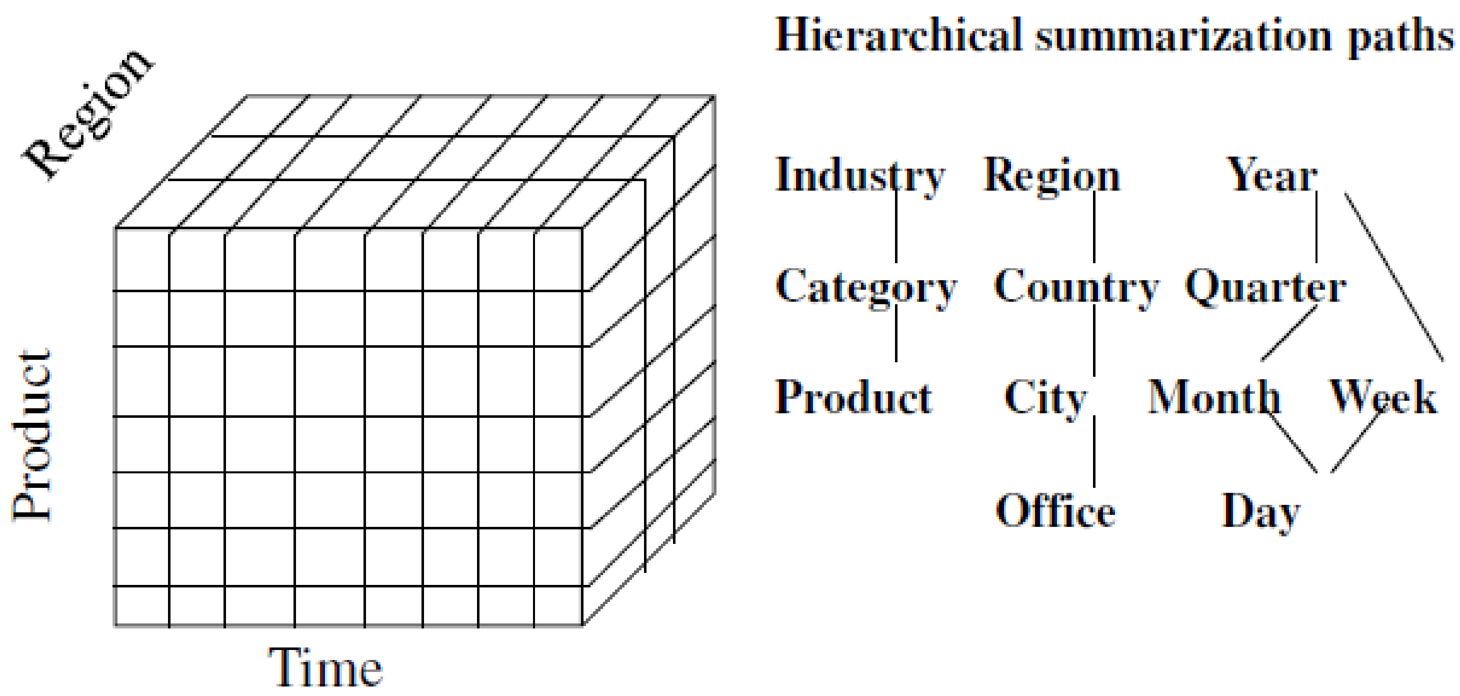


OLAP Operations

OLAP Operations



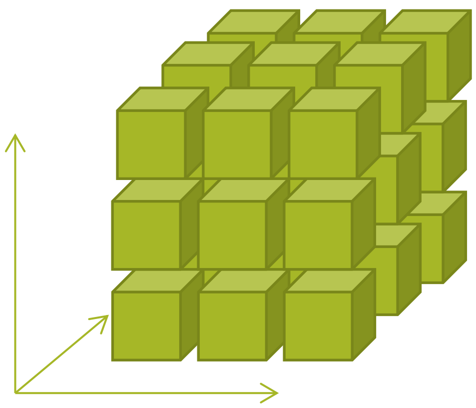
OLAP Operations



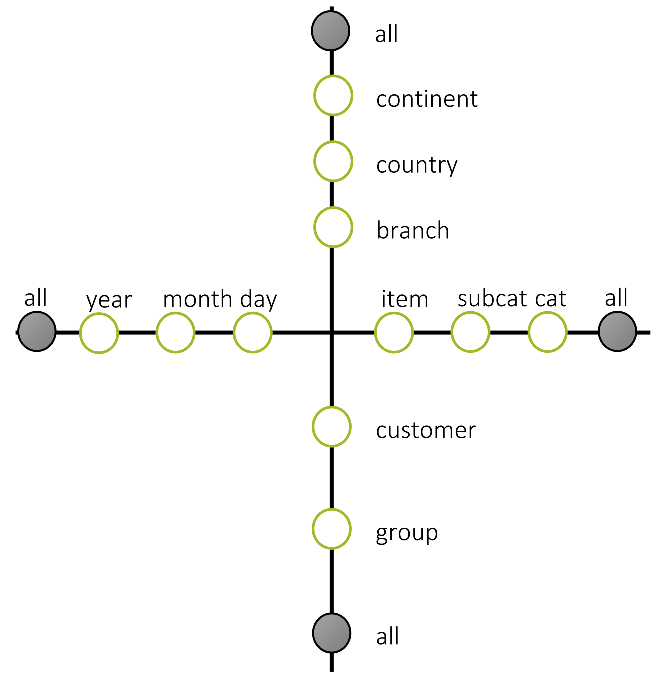
OLAP Operations

Data in a cube

What can we do with the cube ?



Starnet query model



OLAP Operations

Drilling into data

The word *drill* connotes digging deeper into something.

In a dimensional context, that something is a fact.

A generic concept of drilling is expressed simply as *the addition of dimensional detail*.

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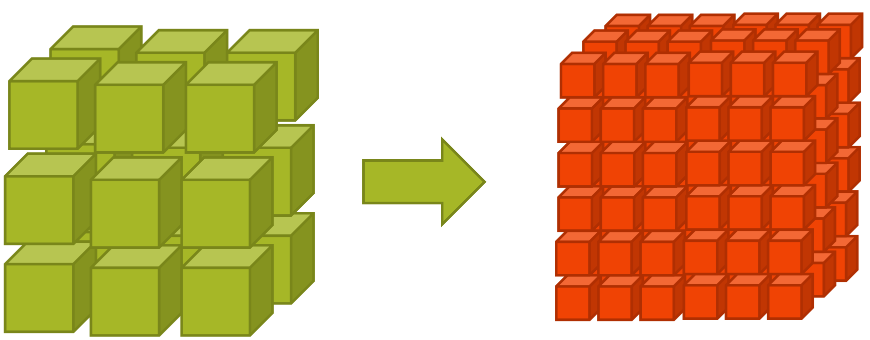
Category	MONTH	Order Dollars
Boxes		88,561.00
Mailers		18,590.23
Tape		143,424.71
Pens		20,997.93

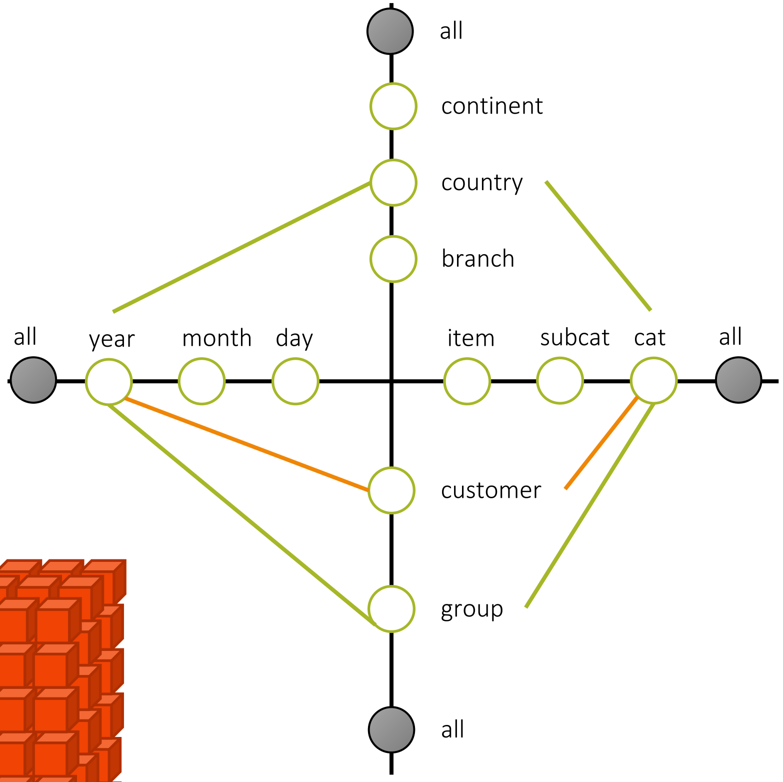
3

Category	Month	Order Dollars
Boxes	January	23,520.20
	February	33,120.03
	March	31,920.77
Mailers	January	8,544.00
	February	7,763.01
	March	2,283.22
Tape	January	49,920.01
	February	29,088.37
	March	64,416.33
Pens	January	14,997.84
	February	2,880.01
	March	3,120.08

OLAP Operations

- OLAP operations
  - Drill down (roll-down)
    - *Category → Product*
    - *Region → City*
    - *Quarter → Month*





OLAP Operations

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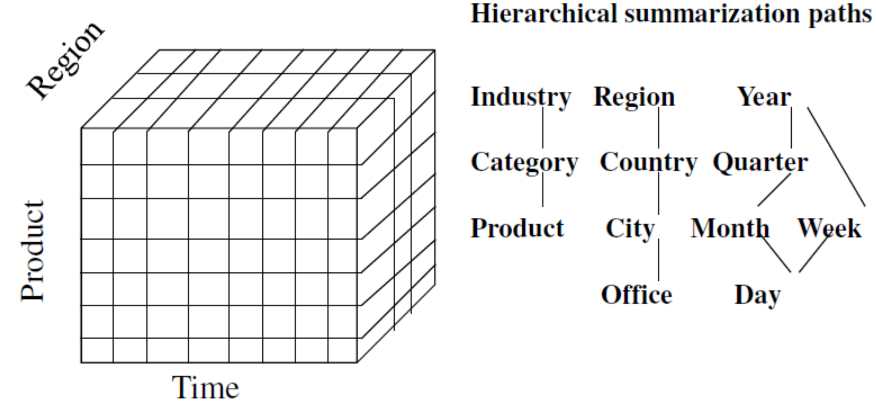
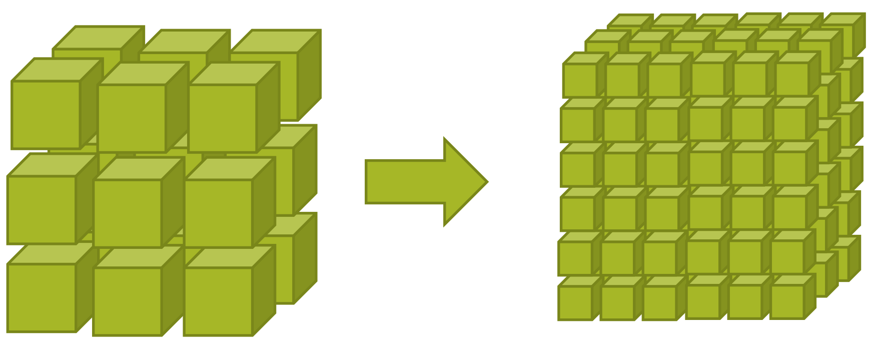
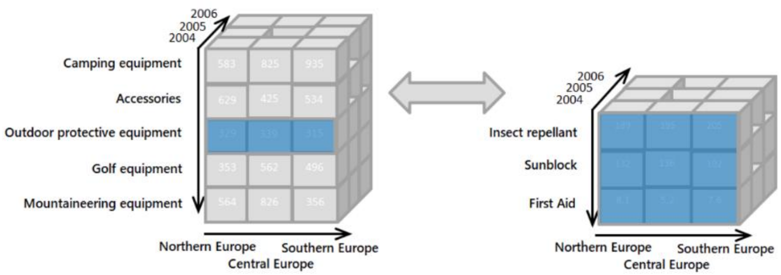
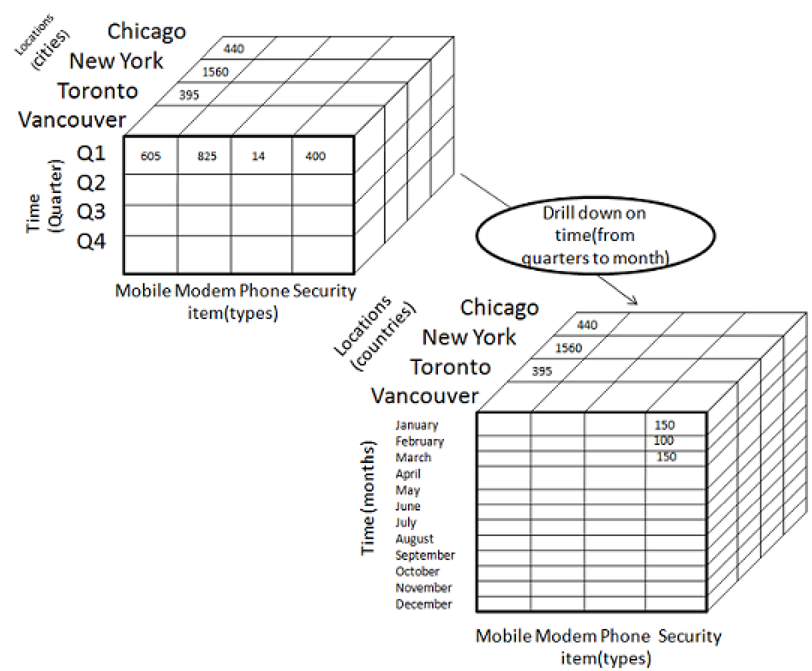




OLAP Operations

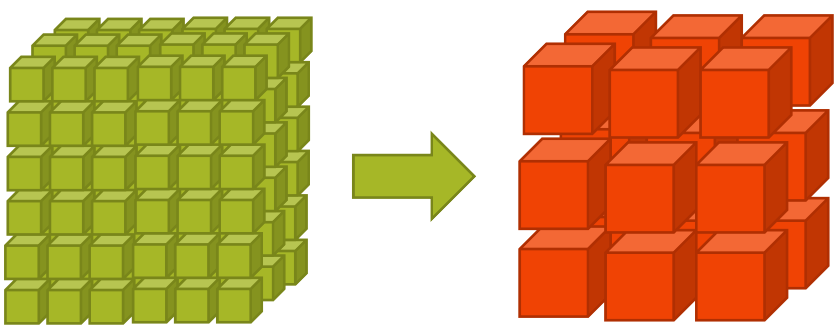


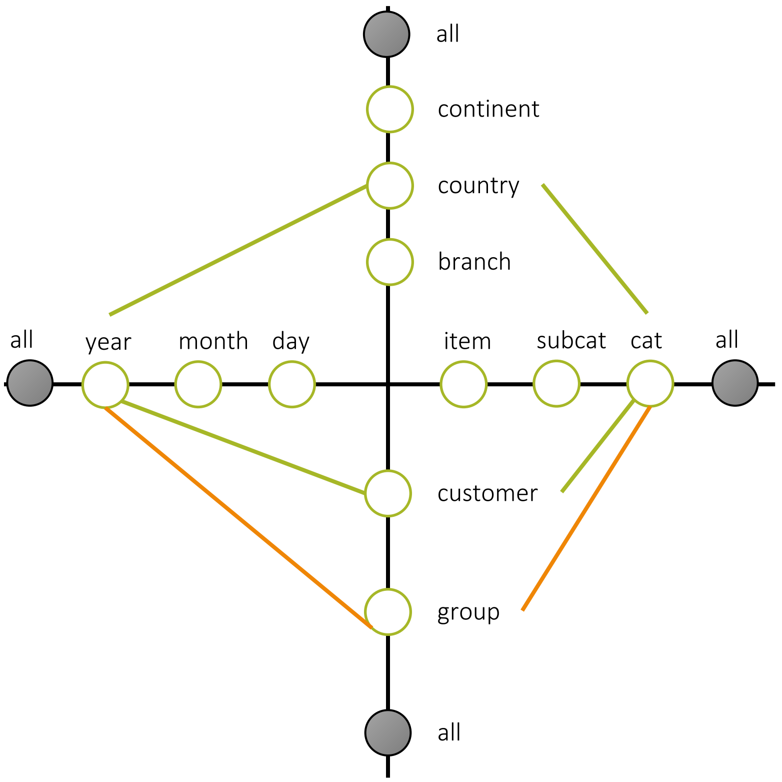


OLAP Operations

OLAP operations

- Roll up (drill-up)
  - *Product → Category*
  - *City → Region*
  - *Month → Quarter*

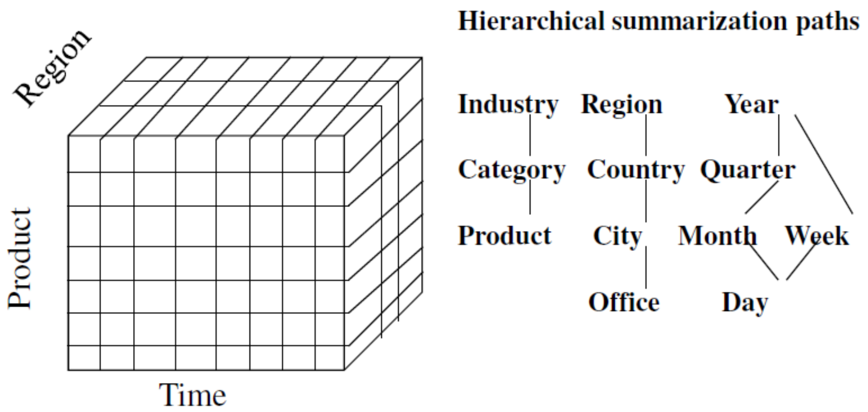


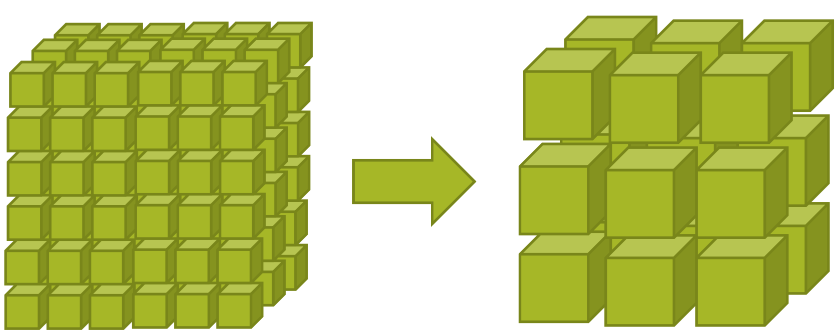


OLAP Operations

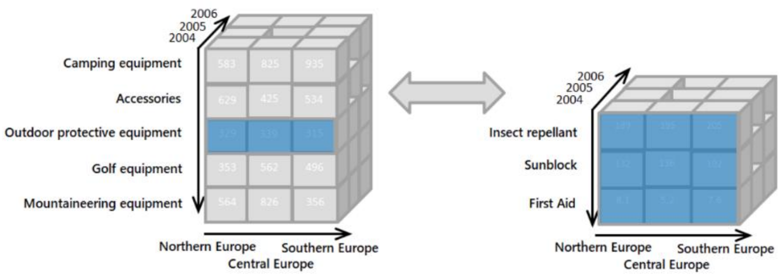
OLAP operations

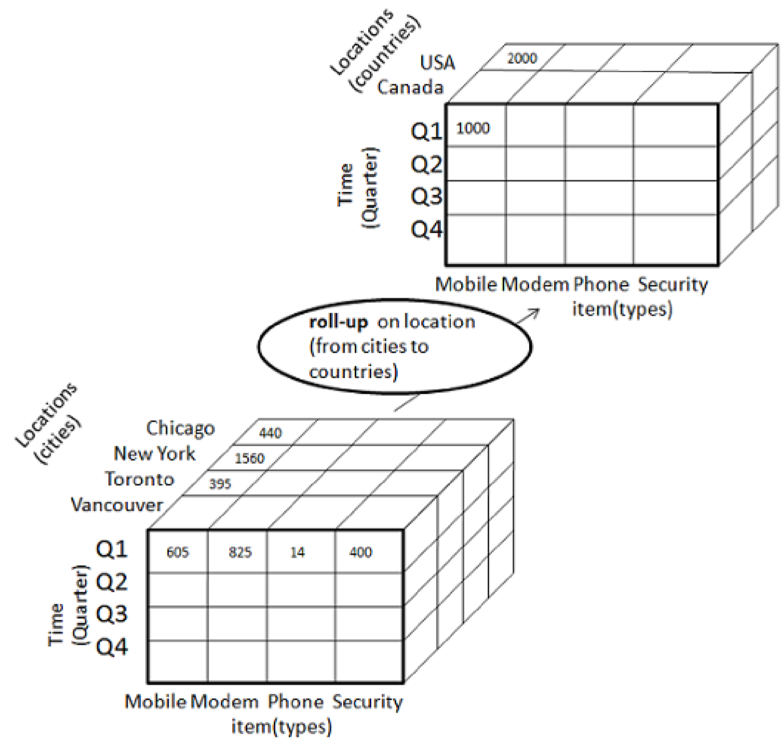
- Roll up (drill-up)
  - *Product → Category*
  - *City → Region*
  - *Month → Quarter*





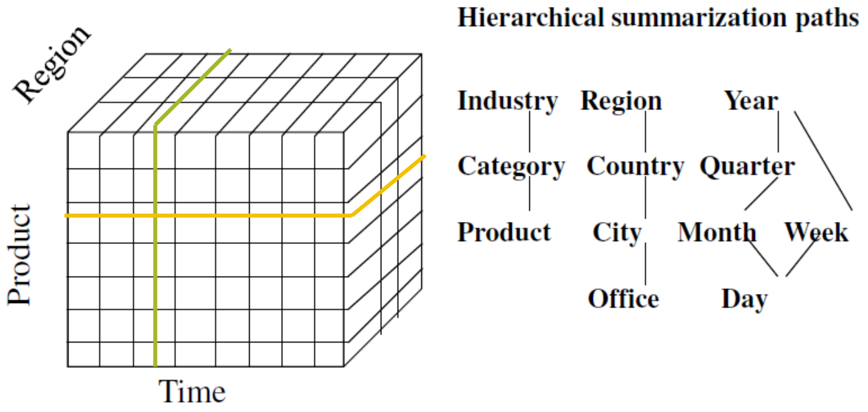
OLAP Operations

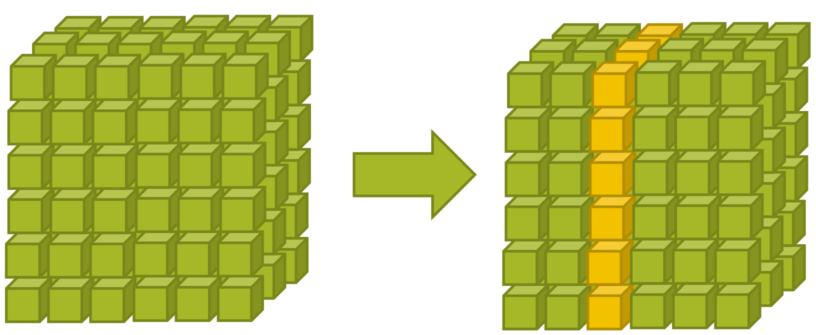




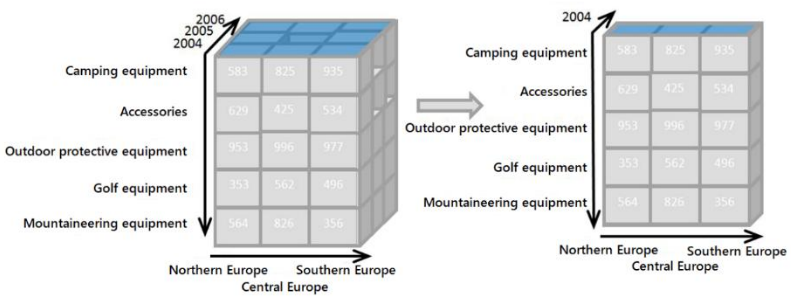
OLAP Operations

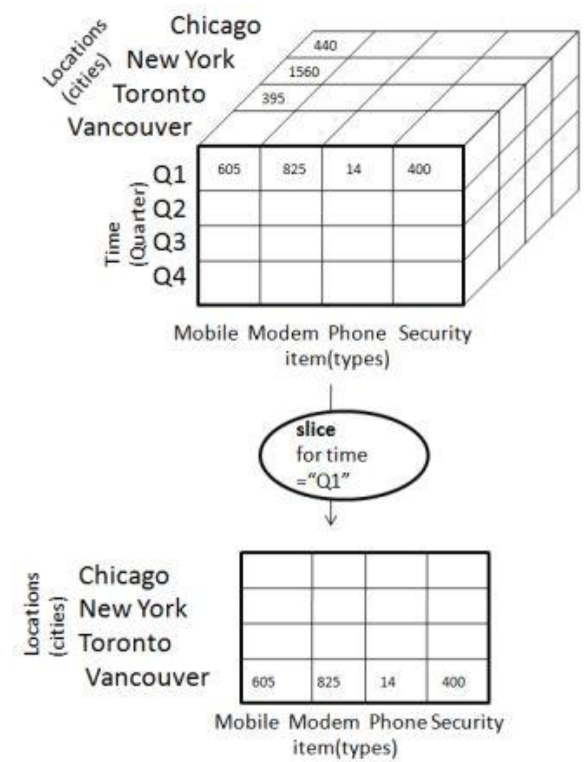
- OLAP operations
  - Slice
    - *Selection on one dimension of the given cube*
    - *Result is a subcube*





OLAP Operations

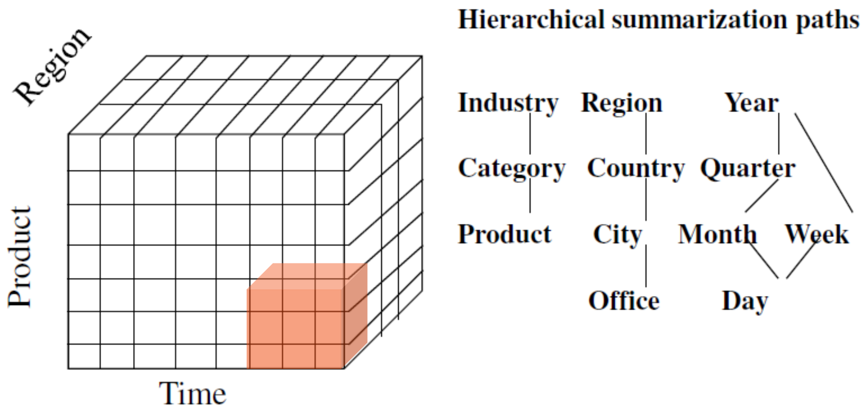


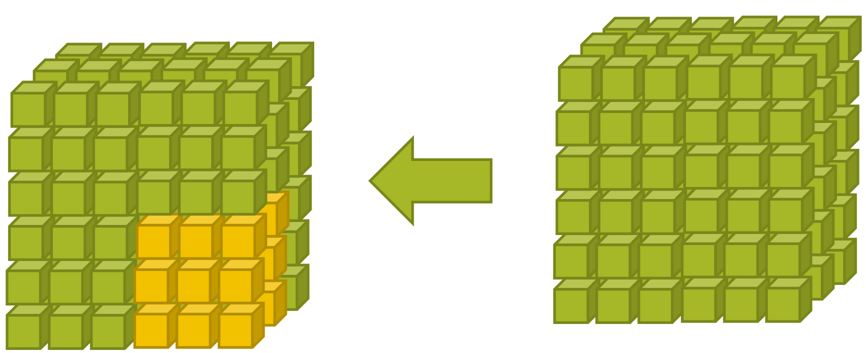


OLAP Operations

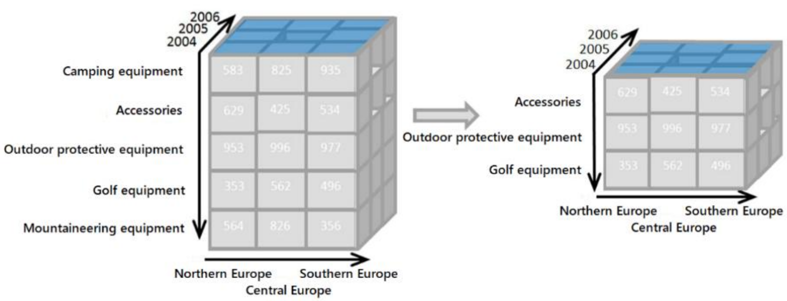
OLAP operations

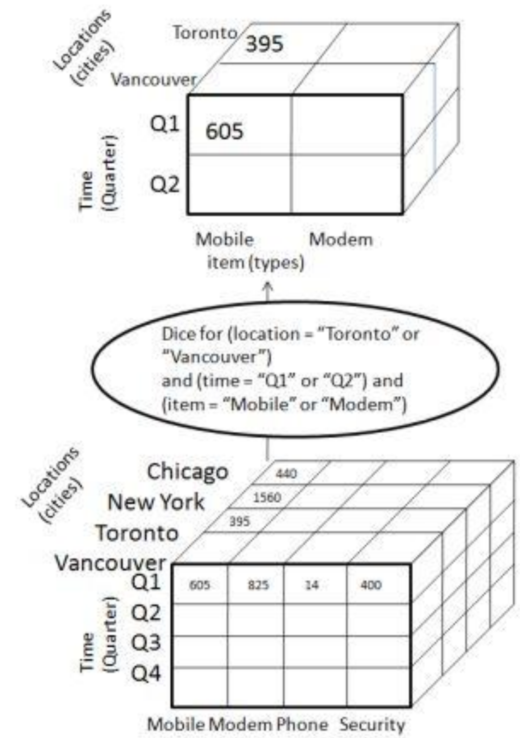
- Dice
  - *Selection on two or more dimensions*
  - *Results in a subcube*





OLAP Operations

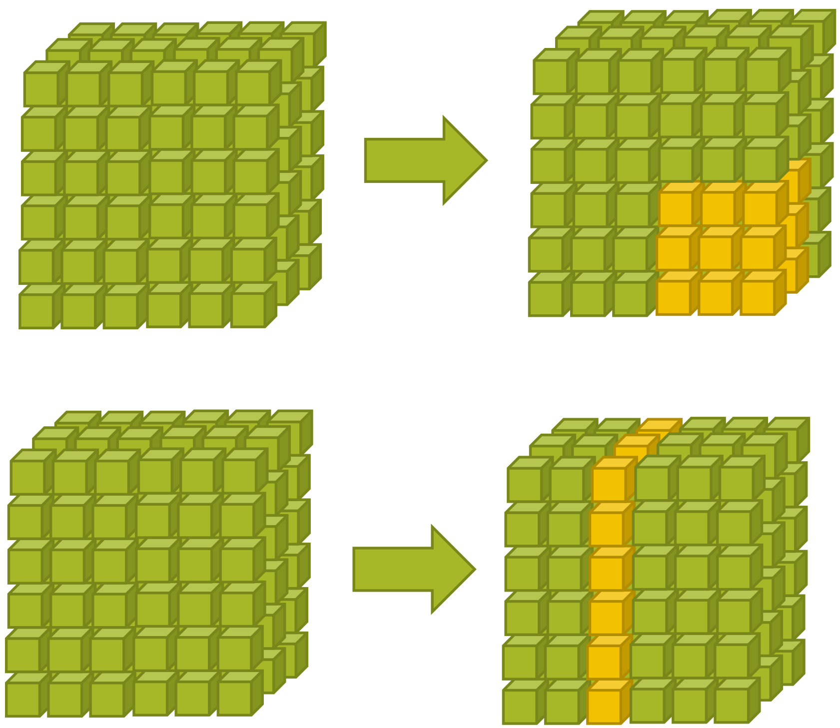


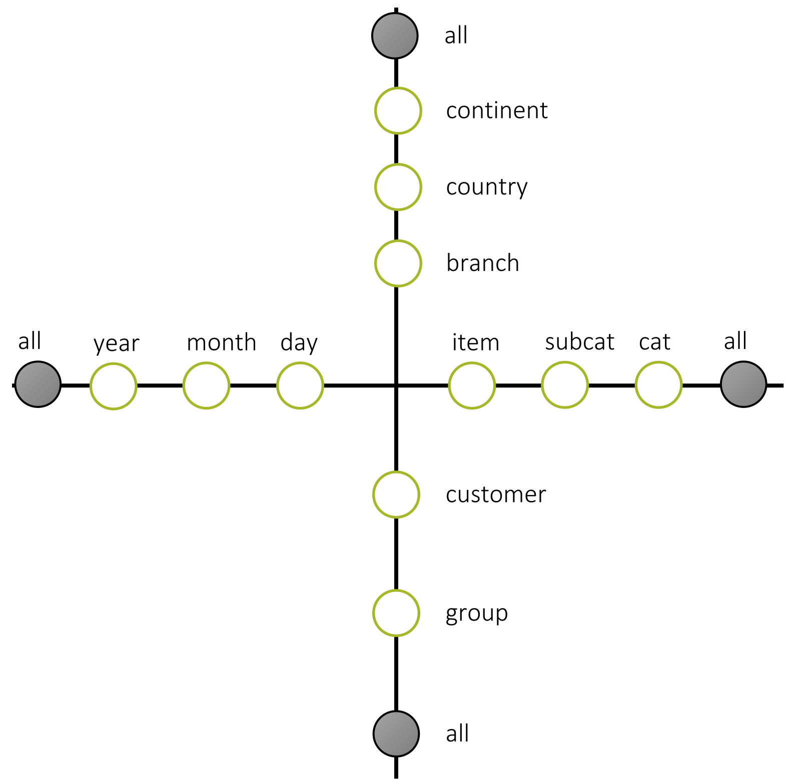


OLAP Operations

OLAP operations

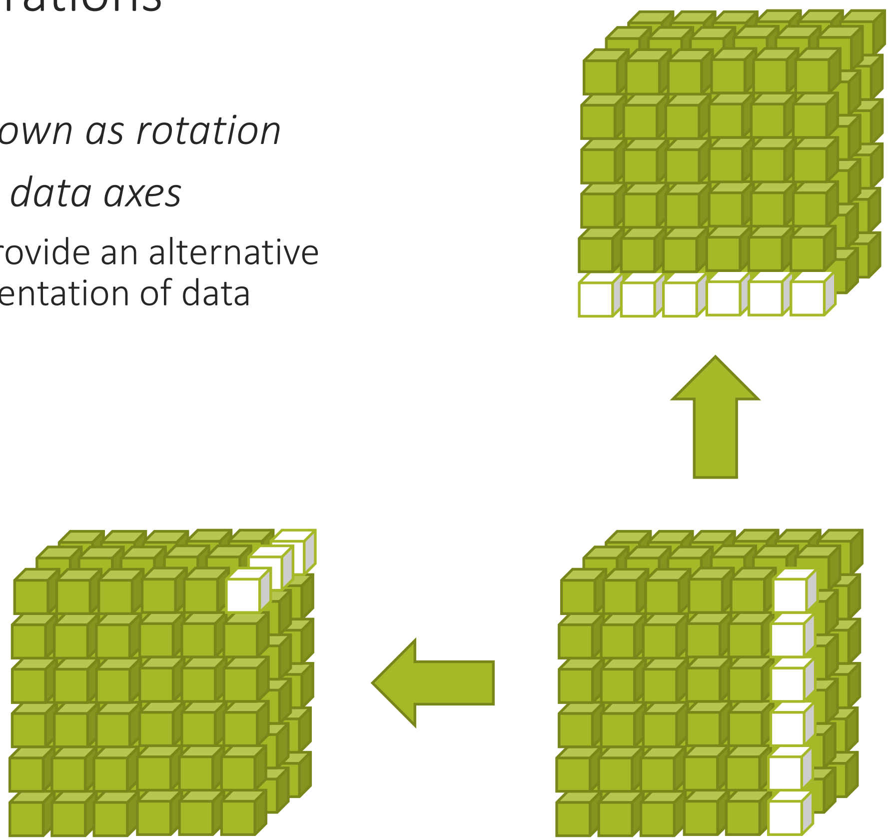
Slice and dice



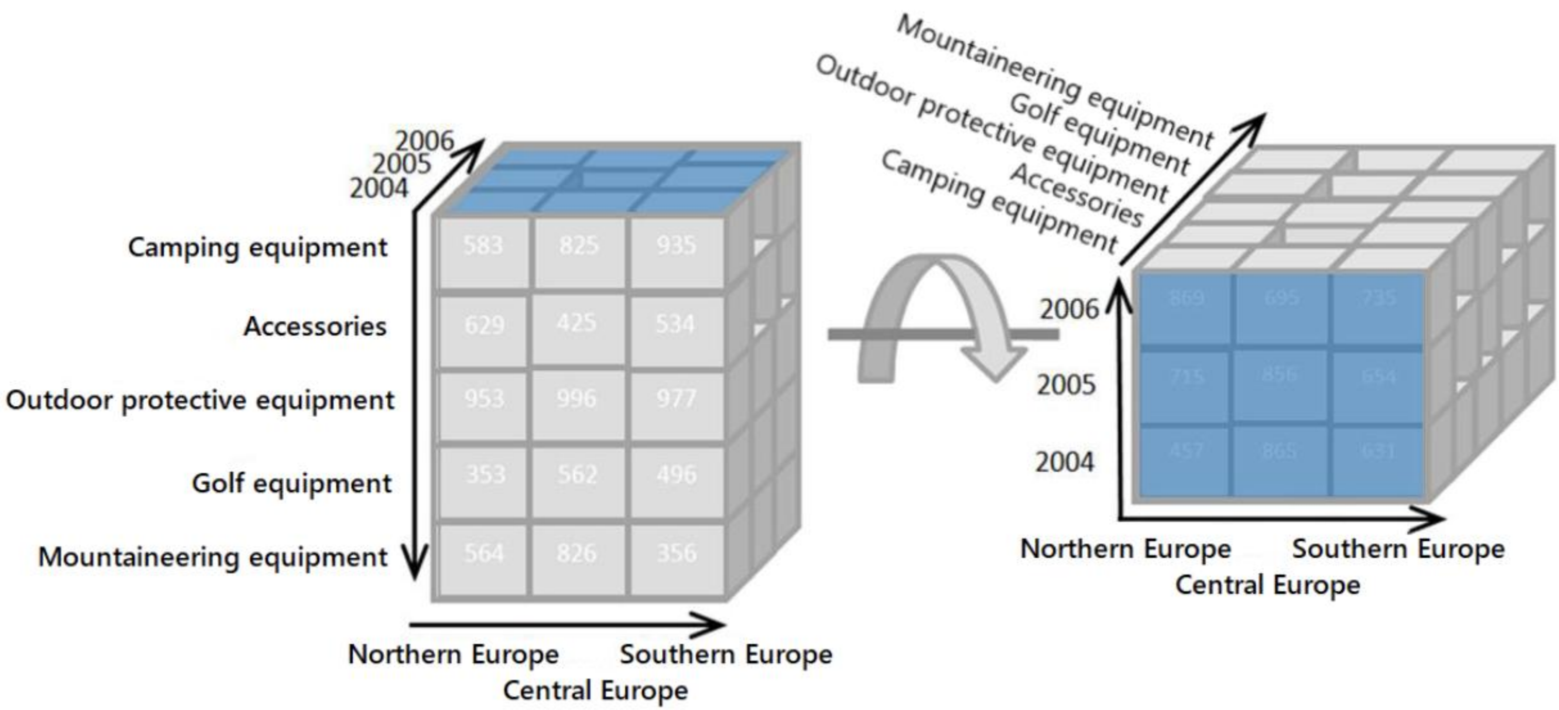


OLAP Operations

- OLAP operations
  - Pivot
    - *also known as rotation*
    - *rotates data axes*
      - to provide an alternative presentation of data



OLAP Operations



OLAP Operations

OLAP operations

- Roll up (drill-up)
  - *summarize data by climbing up hierarchy or by dimension reduction*
  - *It navigates the data from highly detailed data to less detailed data*
- Drill down (roll down)
  - *reverse of roll-up - from higher level summary to lower level summary or detailed data, or introducing new dimensions*
  - *It navigates the data from less detailed data to highly detailed data*
- Slice
  - *project and select*
- Dice
  - *form a new sub cube*
- Pivot (rotate)
  - *reorient the cube, visualization, 3D to series of 2D planes.*

OLAP Operations

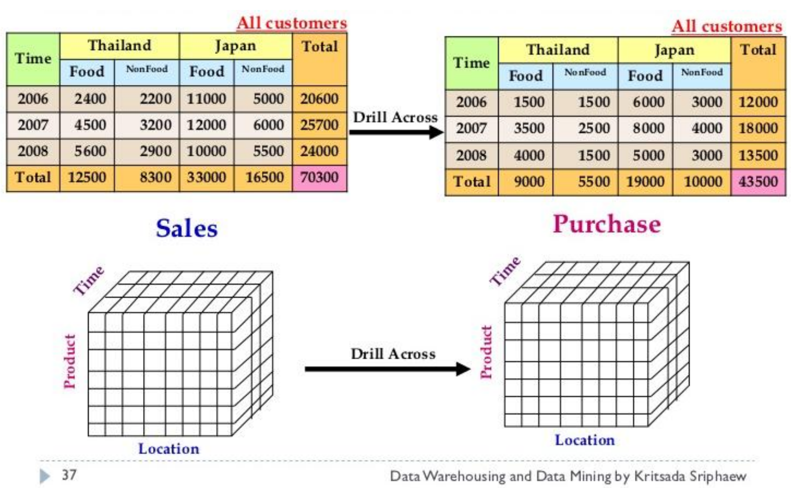
OLAP operation

Drill across

Set together multiple facts

Requires a conformed dimension

executes queries involving (i.e., across) more than one fact table



All customers

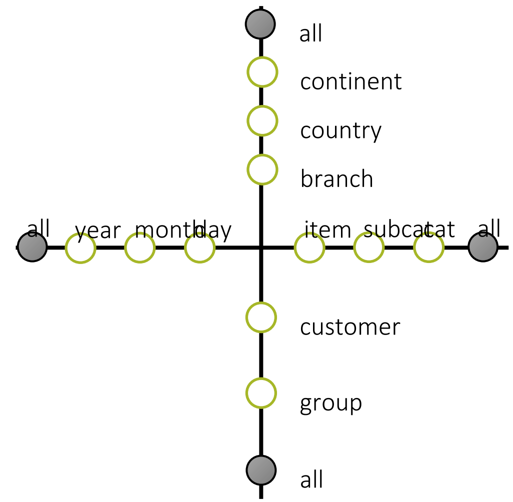
Time	Thailand		Japan		Total
	Food	NonFood	Food	NonFood	
2006	2400	2200	11000	5000	20600
2007	4500	3200	12000	6000	25700
2008	5600	2900	10000	5500	24000
Total	12500	8300	33000	16500	70300

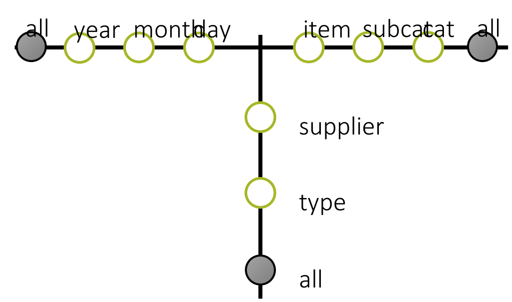
Sales

All customers

Time	Thailand		Japan		Total
	Food	NonFood	Food	NonFood	
2006	1500	1500	6000	3000	12000
2007	3500	2500	8000	4000	18000
2008	4000	1500	5000	3000	13500
Total	9000	5500	19000	10000	43500

Purchase



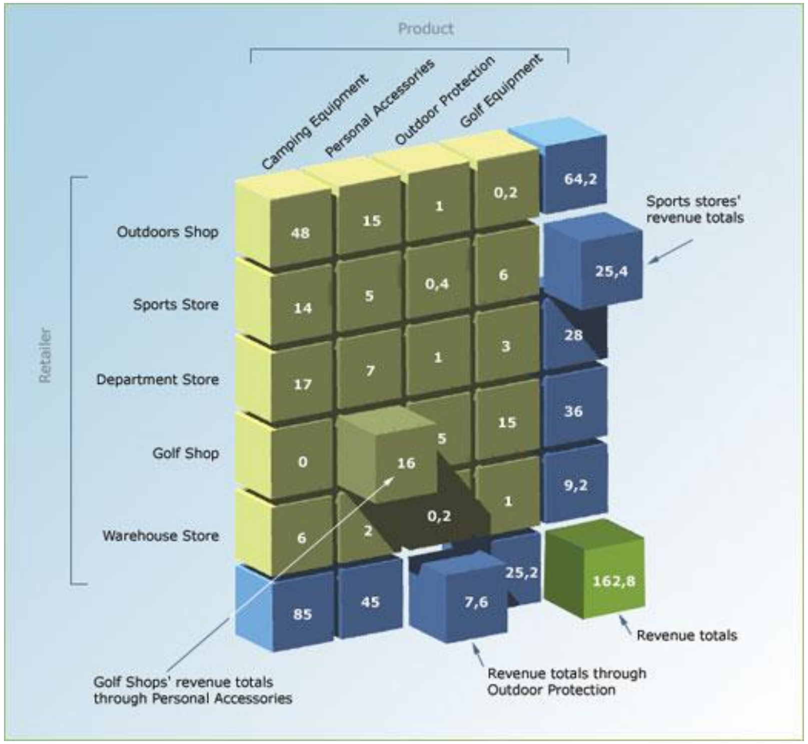


OLAP-oriented Data Model

Data as a cube

Aggregations

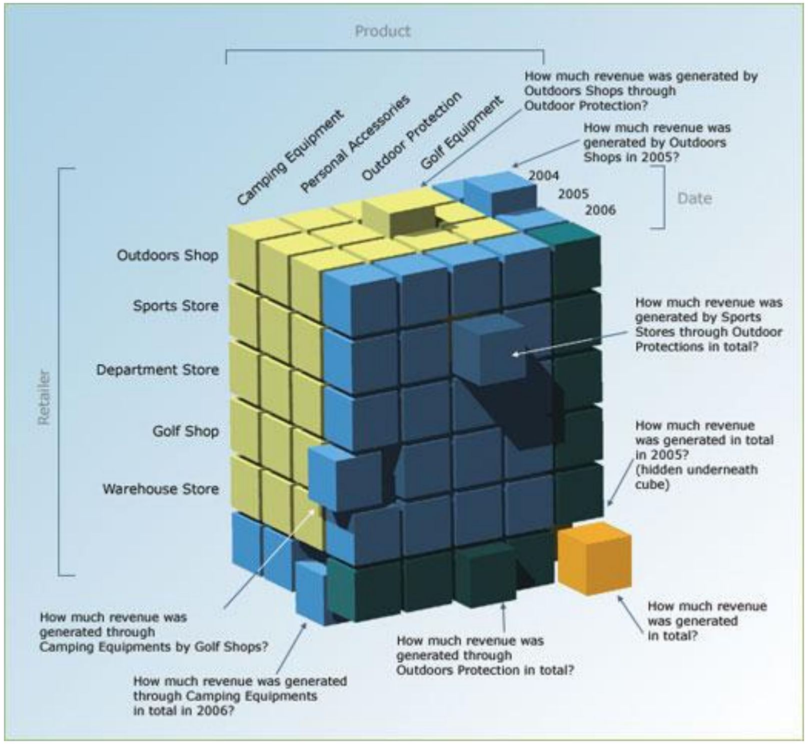
- OLAP queries also require the ability to aggregate data over different dimensions
- We need to have totals



Data as a cube

Aggregations

- Many dimensions = many aggregates
- We need to calculate many different totals



Data as a cube

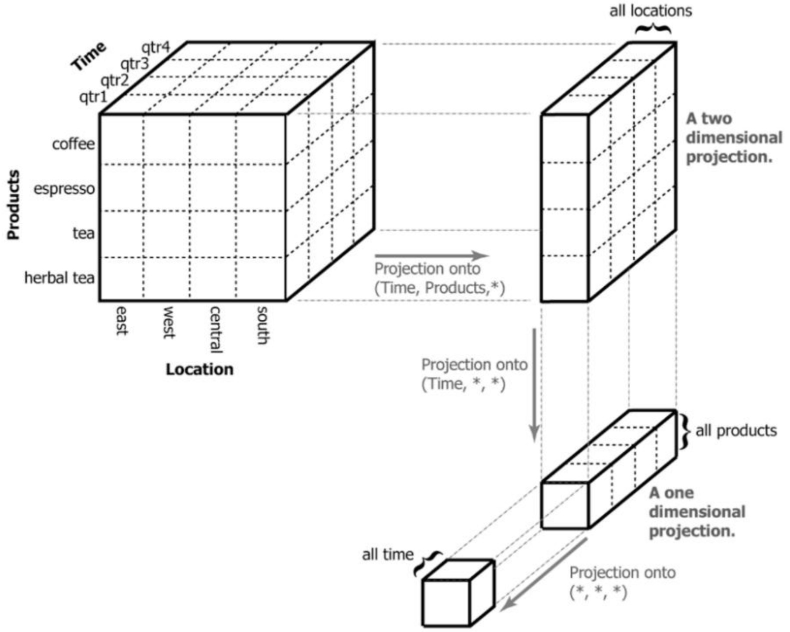
Left to right

How fine-grained data turns into a cube.

Right to left

How the cube containing all the data in one value is broken down:

first by time, then by product, then by location.



Data as a cube

How you do you ask questions of such a dataset?

- You specify how you want it broken down and how to aggregate the measures.

Example

- Dataset of incomes for many different job titles, ages, genders, education levels, etc.
- Ask about the average income for men vs. women

It's the dataset perspective

Process:

- Split the dataset along the gender dimension – into two groups.
- Calculate the averages for each gender.

Data as a cube

This way, you can quickly ask many different questions:

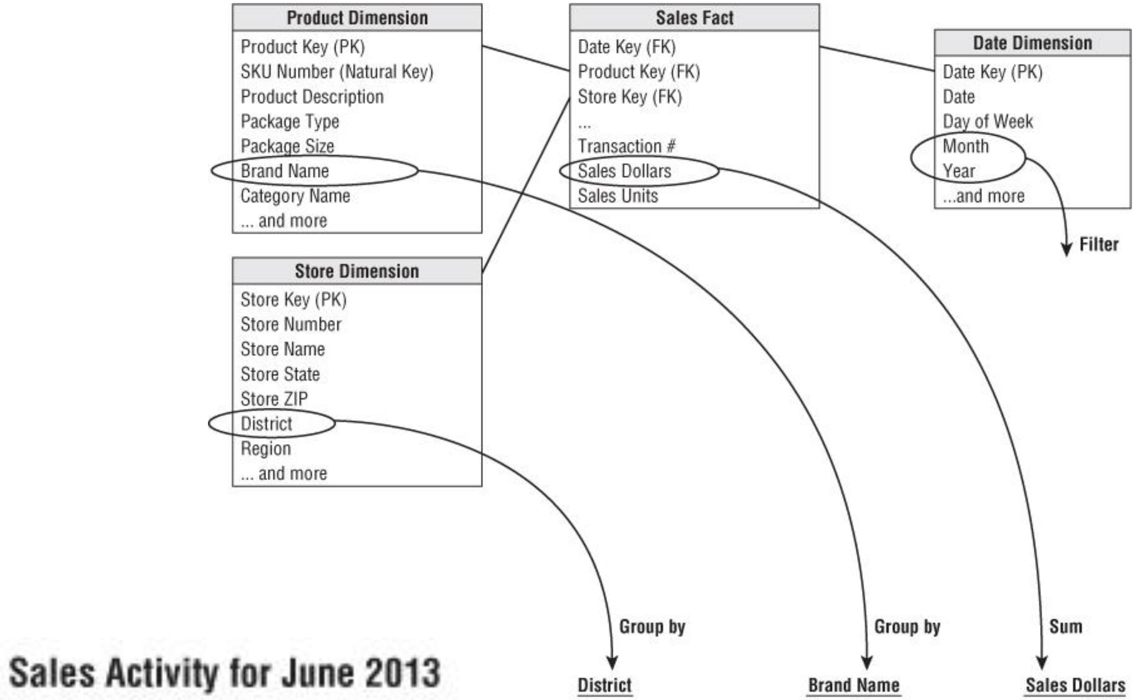
- What's the average by education level?
- How about gender *and* education level?
- What about age?
- Etc.

Different aggregations in each slice are possible

- average, median, minimum, maximum, etc.

It's all just a matter of asking the question.

Analytical system

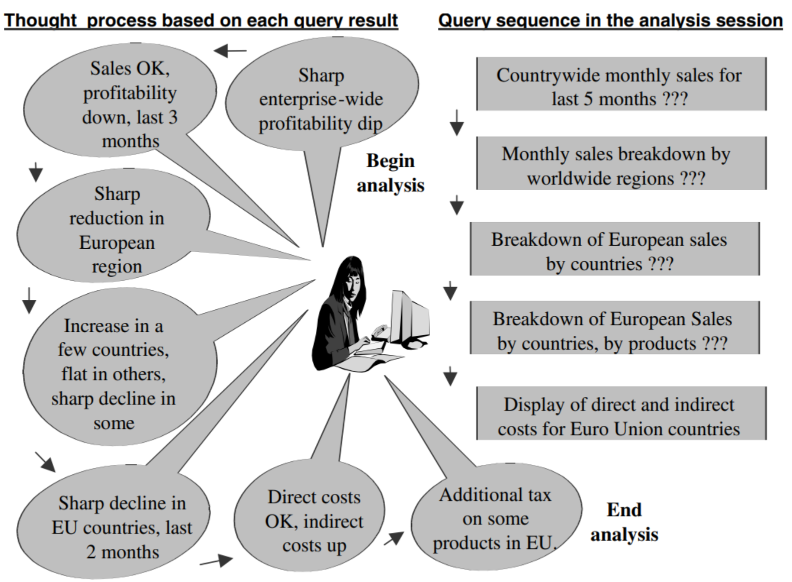


District	Brand Name	Sales Dollars
Atherton	PowerClean	2,035
Atherton	Zippy	707
Belmont	Clean Fast	2,330
Belmont	Zippy	527

Analytical system

The analysis does not stop with this single multidimensional query.

- The user continues to ask further questions
  - *comparisons to similar products,*
  - *comparisons among territories,*
  - *views of the results by rotating the presentation between columns and rows*



Dimensions

Are rarely flat

Dimensions

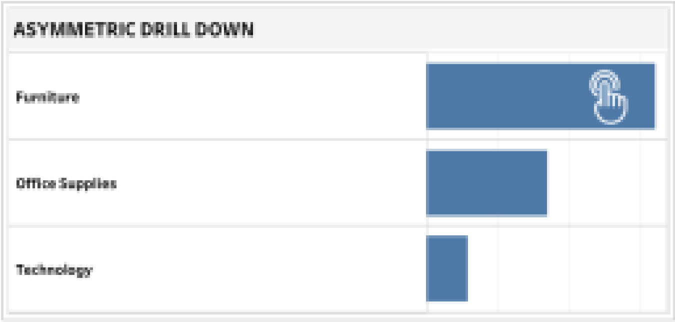
Dimensions are rarely flat

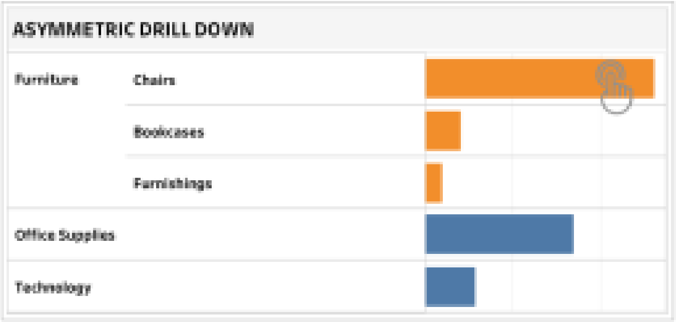
When you look at a report, for example, you might decide you want to know more.

analysis as the process of "drilling into data"

"drill down" - a summarized view is replaced with a more detailed view

This interactive exploration of facts characterizes much of the interaction users have with the data warehouse or data mart.

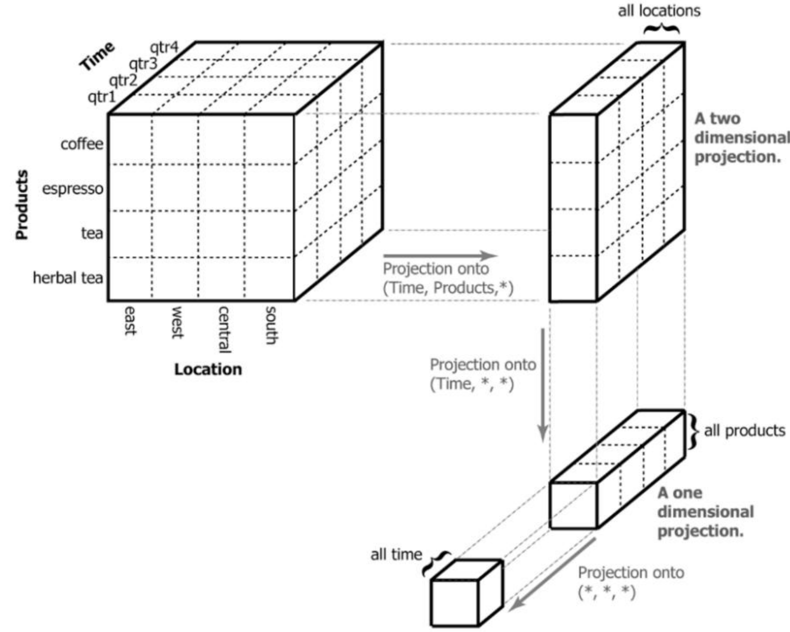




Dimensions

Attribute hierarchies offer a natural way to organize facts at successively deeper levels of detail.

Users understand them intuitively, and drilling through a hierarchy may closely match the way many users prefer to break down key business measurements.



Dimensions

Attribute hierarchies offer a natural way to organize facts at successively deeper levels of detail

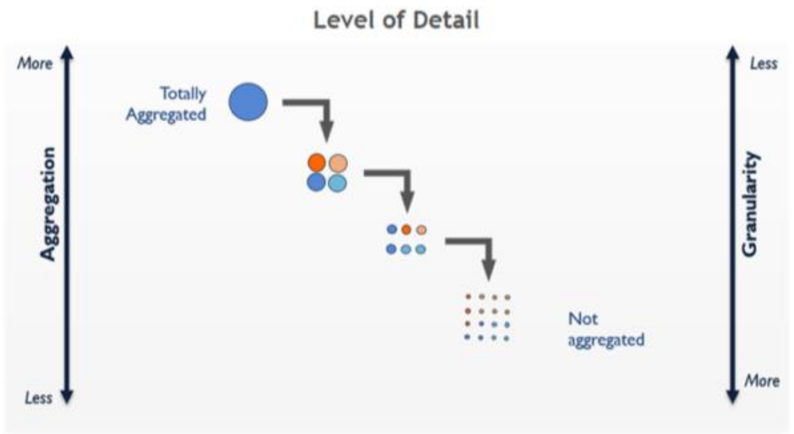
- Users understand them intuitively
  - *drilling through a hierarchy may closely match the way many users prefer to break down key business measurements*

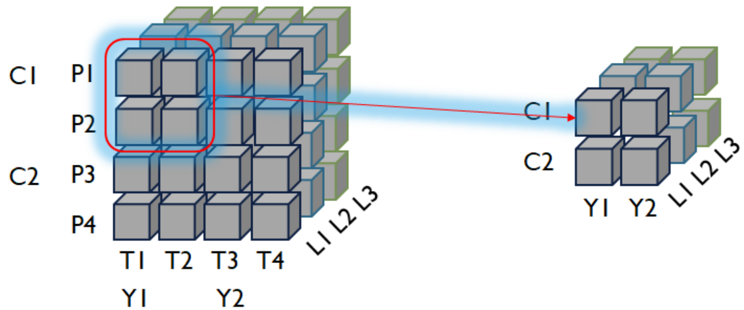
Row Labels	Internet Sales Amount	Reseller Sales Amount	KPI	KPI Goal	KPI Status	KPI Trend
2005	**$3,266,373.66**	**$8,065,435.31**	**11331808.96**	**$9,513,000.00**		
2006	**$6,530,343.53**	**$24,144,429.65**	**30674773.18**	**$29,009,000.00**		
2007	**$9,791,060.30**	**$32,202,669.43**	**41993729.72**	**$38,782,000.00**		
2008	**$9,770,899.74**	**$16,038,062.60**	**25808962.34**	**$18,410,000.00**		
2008 Q1	**$4,283,629.96**	**$7,102,685.11**	**11386315.07**	**$8,051,000.00**		
2008 Jan	$1,340,244.95	$1,662,547.32	3002792.274	$8,051,000.00		
2008 Feb	$1,462,479.83	$2,700,766.80	4163246.633			
2008 Mar	$1,480,905.18	$2,739,370.98	4220276.163			
2008 Q2	**$5,436,429.15**	**$8,935,377.49**	**14371806.64**	**$10,359,000.00**		
2008 Q3	**$50,840.63**		**50840.63**			
2010						
Grand Total	**$29,358,677.22**	**$80,450,596.98**	**109809274.2**	**$95,714,000.00**		

Dimensions

Many dimensions can be understood as a hierarchy of attributes, participating in successive master–detail relationships:

- the bottom of such a hierarchy represents the lowest level of detail
  - *Highest granularity of data*
- the top represents the highest level of summarization
  - *Lowest granularity of data*
- each level may have a set of attributes and participates in a parent–child relationship with the level beneath it.





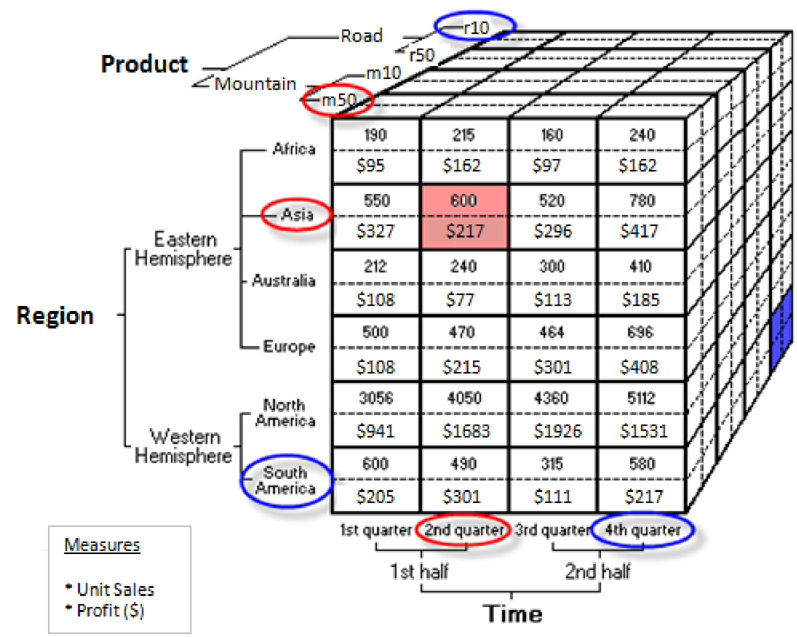
Dimensions

Many dimensions can be understood as a hierarchy of attributes, participating in successive master–detail relationships.

- the bottom of such a hierarchy represents the lowest level of detail
- the top represents the highest level of summarization

Each level may have a set of attributes and participates in a parent–child relationship with the level beneath it.

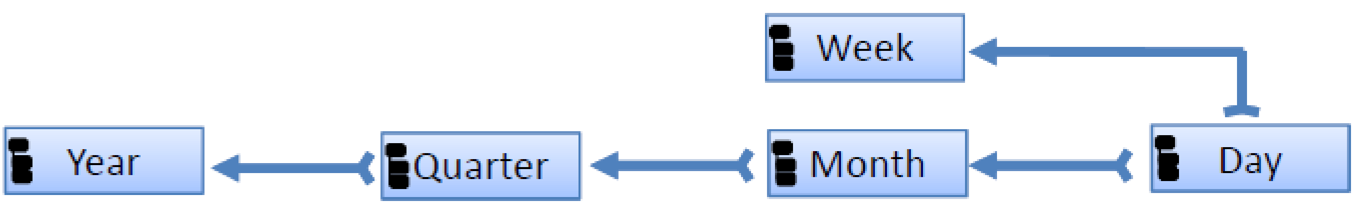
- Days make up months, months fall into quarters, etc.



Dimensions

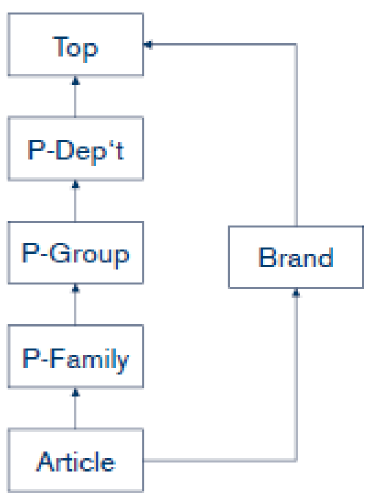
Dimension hierarchies

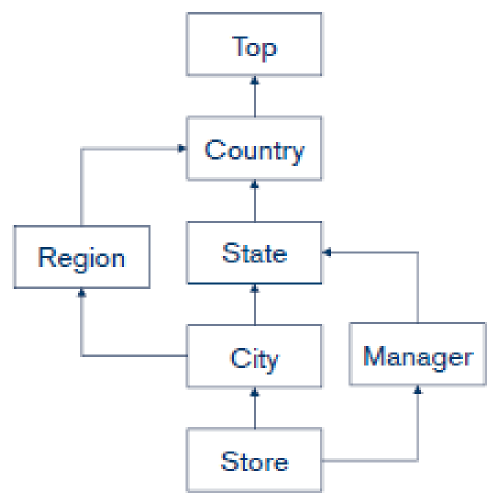
- are organized in classification levels
  - *e.g., Day, Month, …*
- dependencies between the classification levels
  - *are described by the classification schema through functional dependencies*



Dimensions

In general - partially ordered set of dimensional attributes
- ({D1, …, Dn, TopD}; →)
  - → *functional dependency*
    - attribute A determines B (A → B), if the value of B is uniquely determined by the value of A
  - *TopD is the maximum element regarding →*
    - $\forall_{i=1,n}: D_i$→TopD
- Partial ordering allows for parallel hierarchies
- Fully-ordered set of classification levels is called a Path

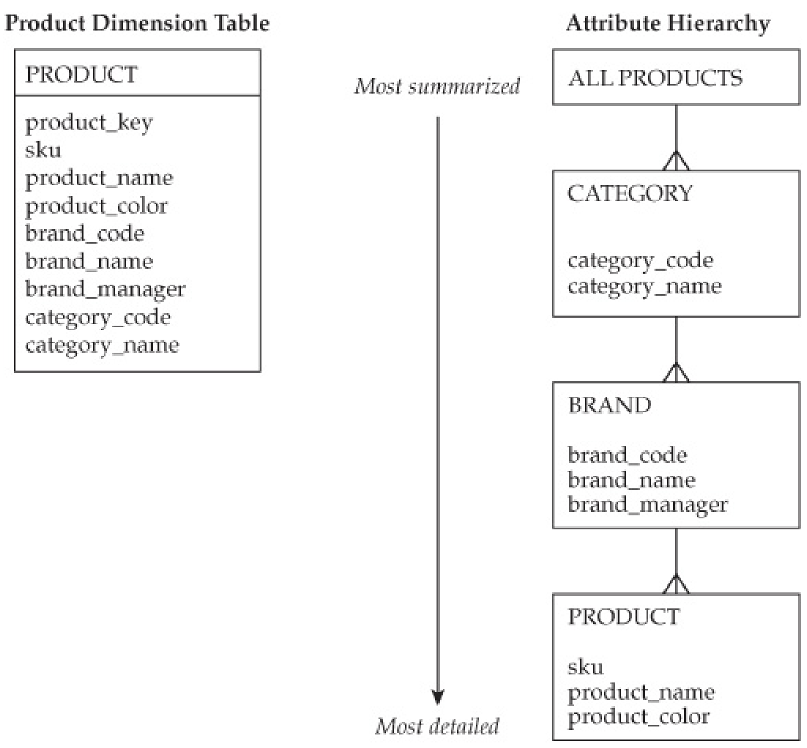




Dimensions

The highest level of summarization, all products, is added for convenience.

- It represents a complete summarization of the product dimension
- studying a fact by all products results in a single row of data

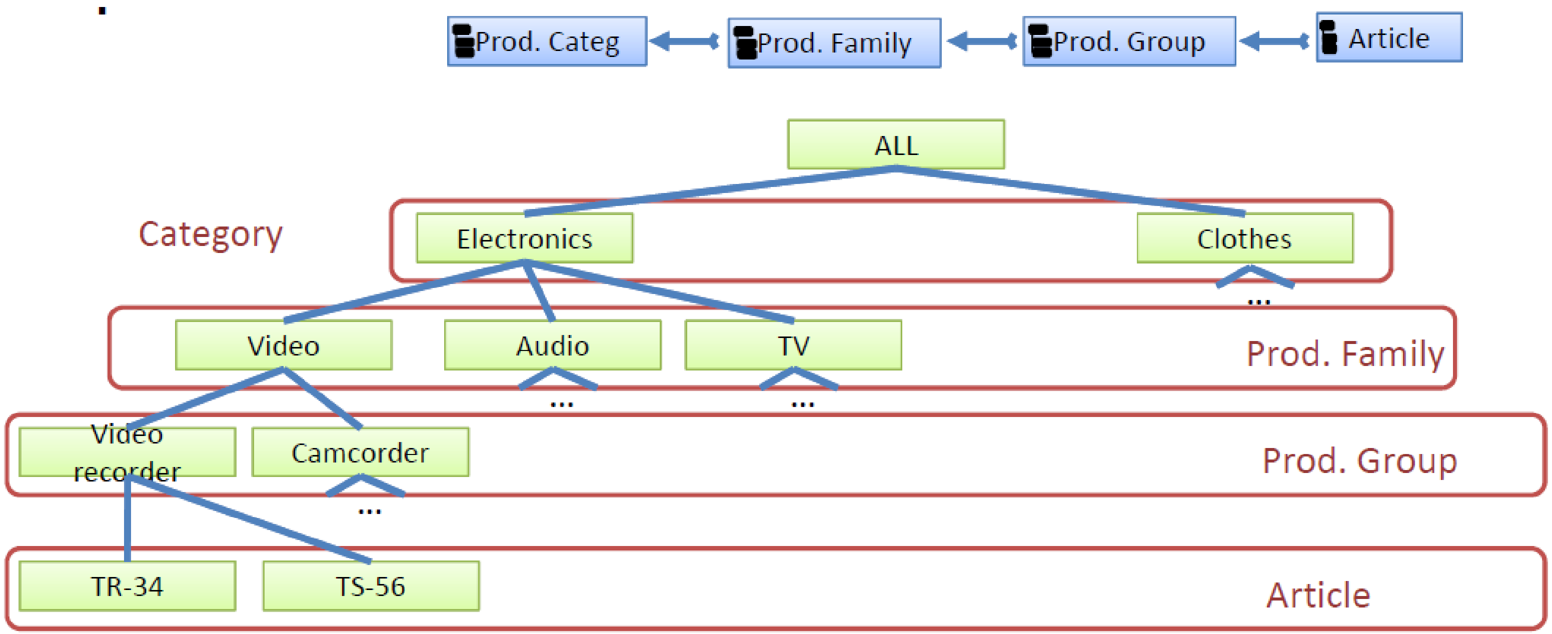


Dimensions

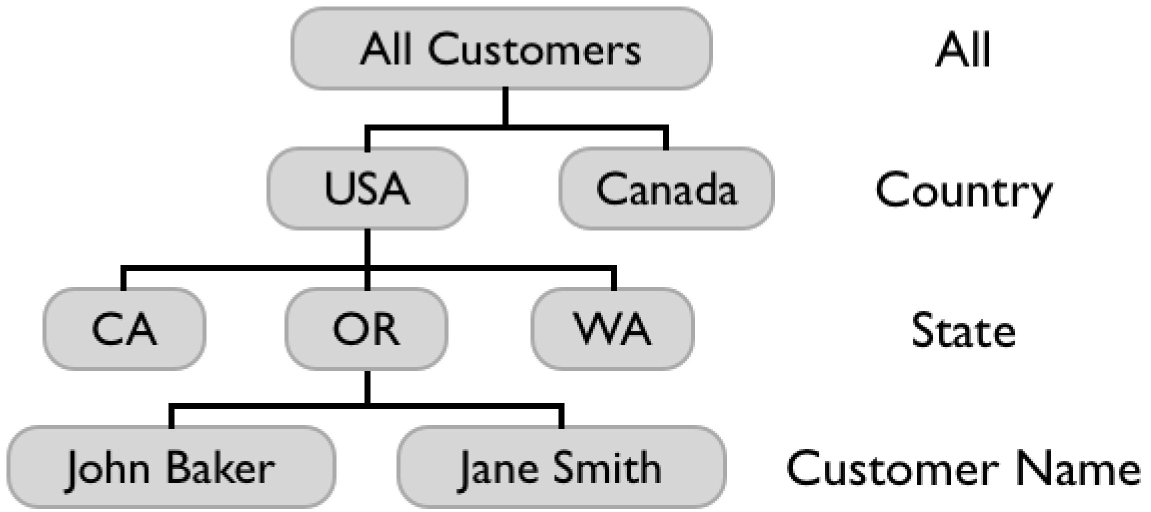
Example:

classification hierarchy path from product dimension

fully-ordered set of classification levels is called a Path



Dimensions


All Customers
USA
Canada
CA
OR
WA
John Baker
Jane Smith
All
Country
State
Customer Name

Dimensions

Some software tools link the concept of drilling to the concept of an attribute hierarchy.

- hierarchy as a predefined drill path
- sometimes referred to as drilling within an attribute hierarchy

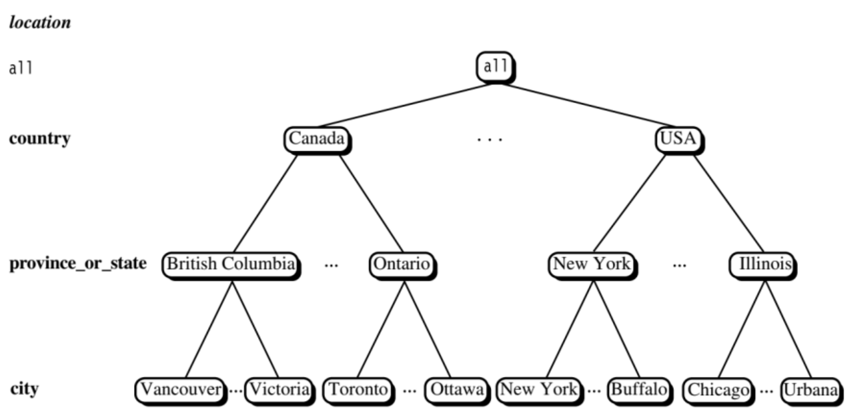
When viewing a fact:

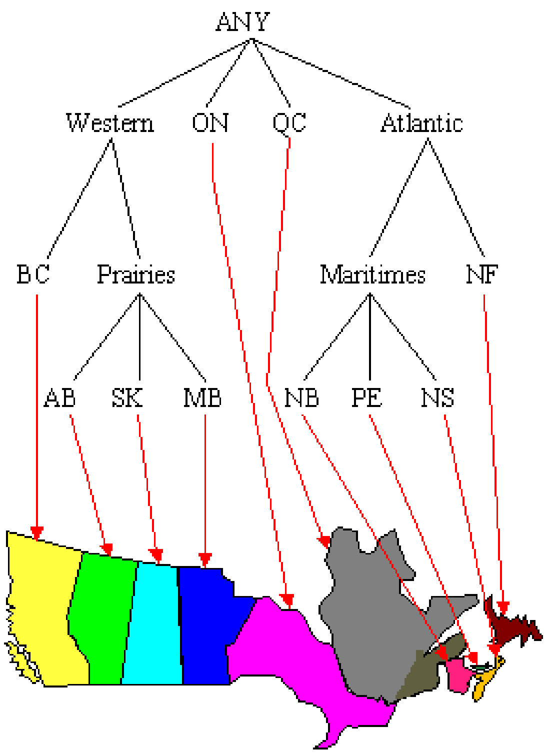
- drilling down
  - *is accomplished by adding a dimension attribute from the next level down the hierarchy*
  - *specifying more detailed context*
    - E.g. by different country state (detailed for each country state)
- drilling up
  - *is achieved by removing a dimension attribute*
  - *Specifying a more general context*
    - E.g. by all country states (aggregated without different country state)

Dimensions

Dimension hierarchies

A **concept hierarchy** defines a sequence of mappings from a set of low-level concepts to higher-level, more general concepts.

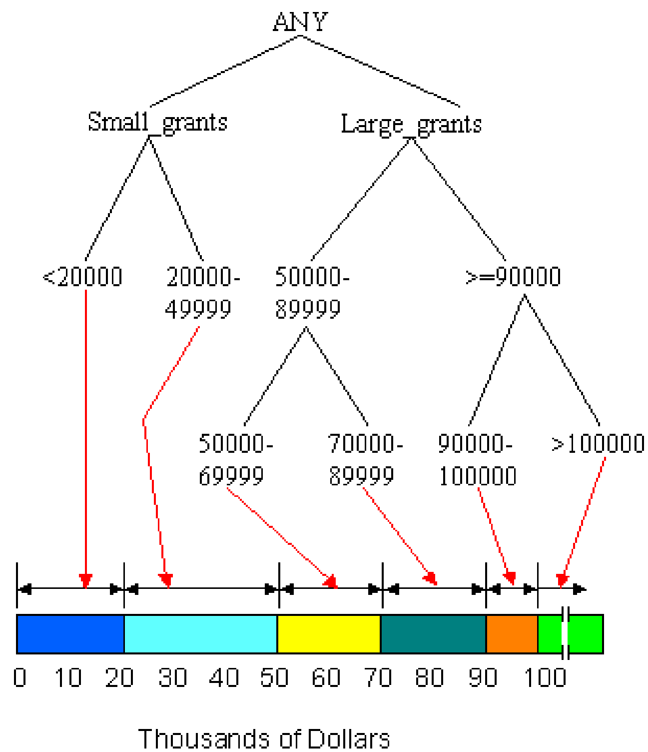


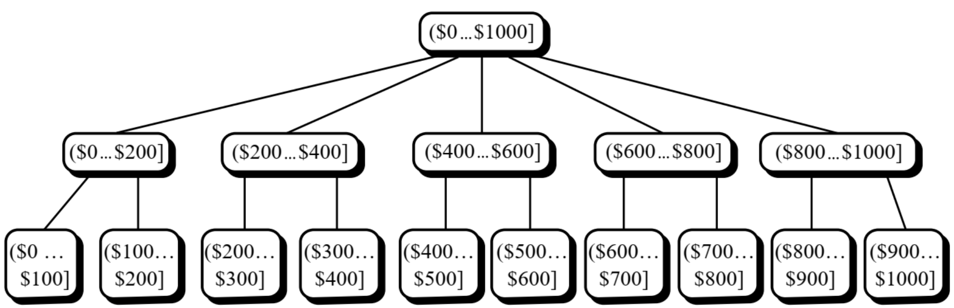


Dimensions

Dimension hierarchies

may also be defined by discretizing or grouping values for a given dimension or attribute, resulting in a **set-grouping hierarchy**

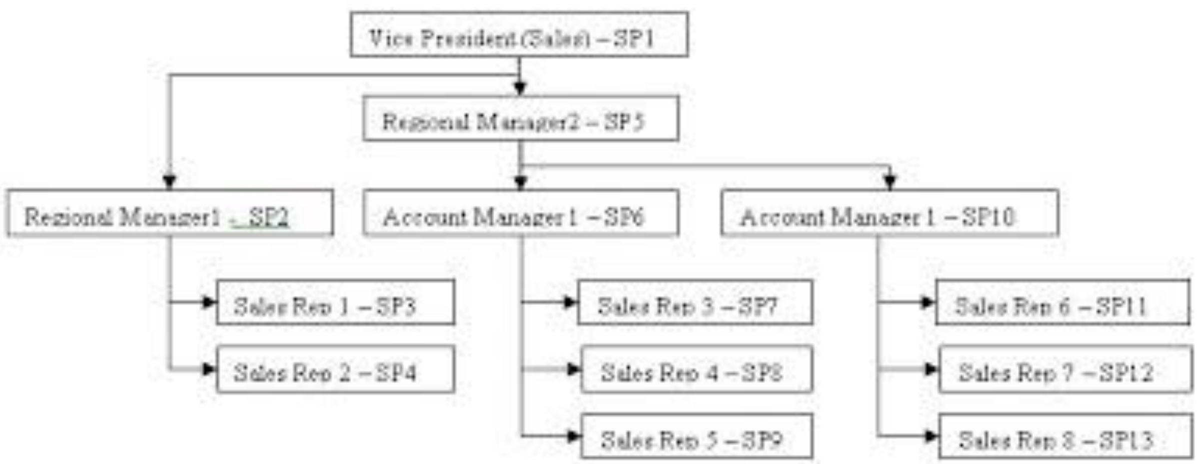




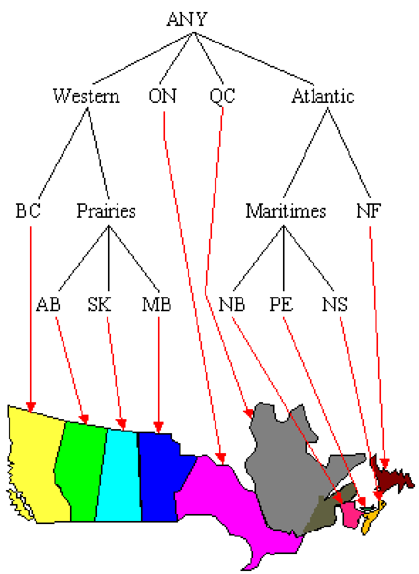
Dimensions

Instance hierarchy may be useful in studying facts

- suppose each call has an assigned employee
- employees have managers that they report to
- It may be useful to roll up all transactions to regional managers level
  - *explore the data by drilling down through multiple levels*



Using hierarchies



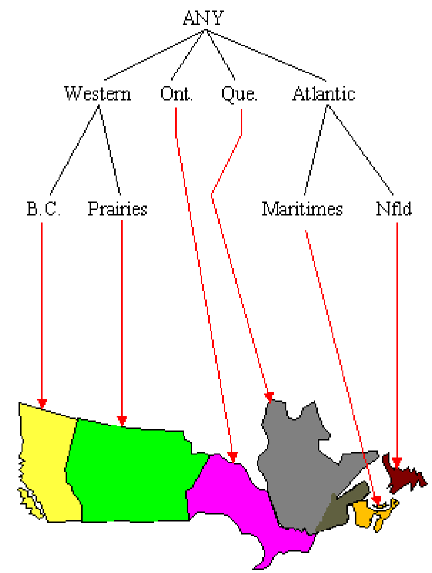
Province	Grant_Amount	Votes
B.C.	>100000	4
B.C.	90000-100000	8
B.C.	70000-89999	34
B.C.	50000-69999	41
Ont.	>100000	9
Ont.	90000-100000	13
Ont.	70000-89999	53
Ont.	20000-49999	234
N.B.	20000-49999	38
N.S.	<20000	15
P.E.I.	20000-49999	6
P.E.I.	<20000	65
Alb.	90000-100000	4
Alb.	<20000	21
Man.	<20000	112
Sask.	<20000	98
Que.	>100000	12
Que.	90000-100000	43
Que.	70000-89999	65

Generalize

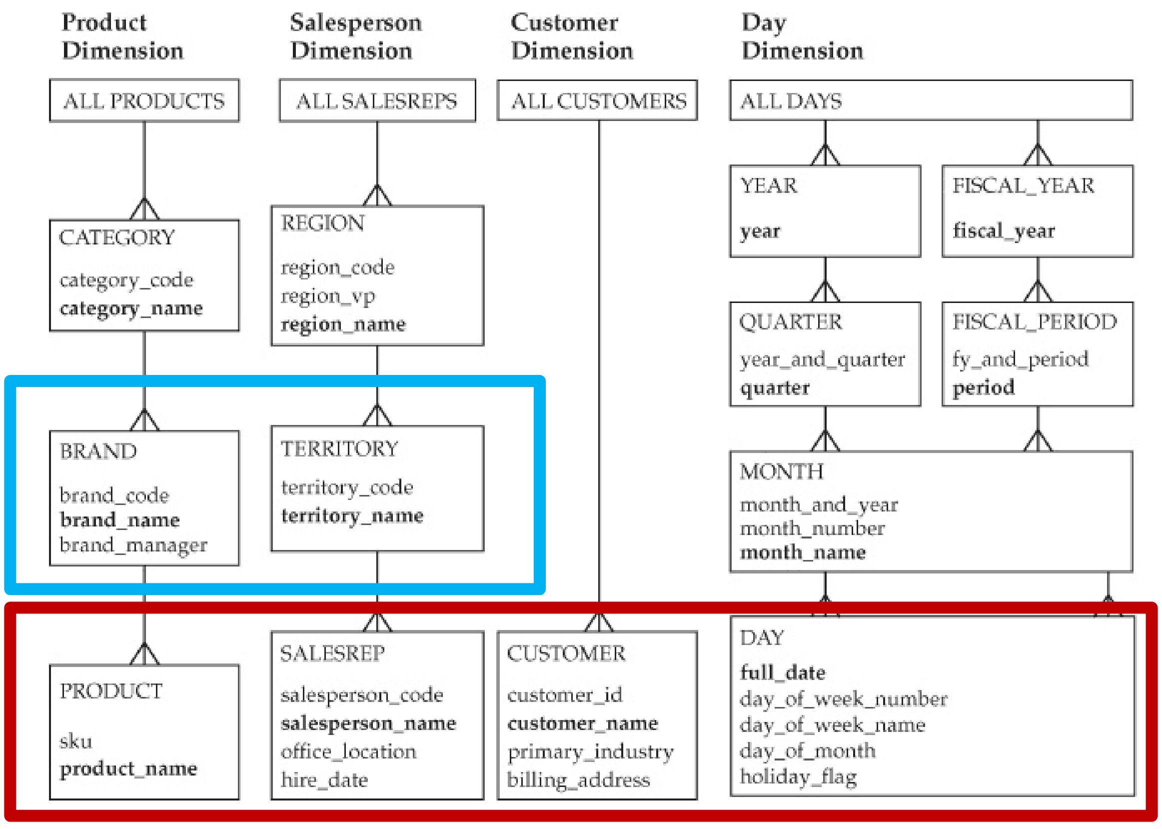
Maritimes	20000-49999	38
Maritimes	<20000	15
Maritimes	20000-49999	6
Maritimes	<20000	65

Generalize

Prairies	90000-100000	4
Prairies	<20000	21
Prairies	<20000	112
Prairies	<20000	41



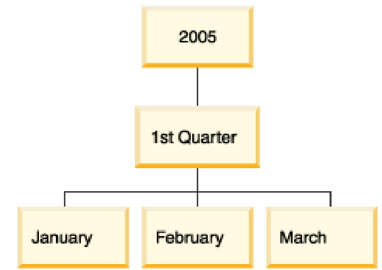
Dimensions

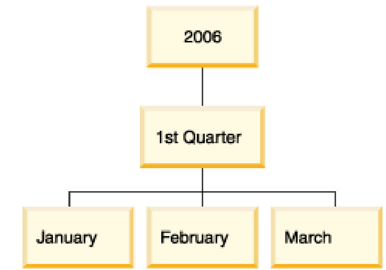


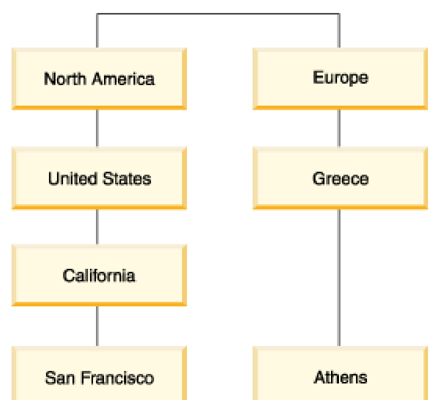
Product Dimension
Salesperson Dimension
Customer Dimension
Day Dimension
ALL PRODUCTS
ALL SALESREPS
ALL CUSTOMERS
ALL DAYS
YEAR
year
FISCAL_YEAR
fiscal_year
CATEGORY
category_code
category_name
REGION
region_code
region_vp
region_name
QUARTER
year_and_quarter
quarter
FISCAL_PERIOD
fy_and_period
period
BRAND
brand_code
brand_name
brand_manager
TERRITORY
territory_code
territory_name
MONTH
month_and_year
month_number
month_name
PRODUCT
sku
product_name
SALESREP
salesperson_code
salesperson_name
office_location
hire_date
CUSTOMER
customer_id
customer_name
primary_industry
billing_address
DAY
full_date
day_of_week_number
day_of_week_name
day_of_month
holiday_flag

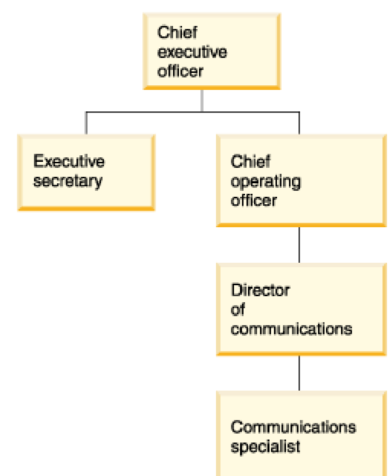
Dimensions

- A hierarchy is a
  - many-to-one relationship between members of a table or between tables
  - basically, consists of different levels, each corresponding to a dimension attribute.
  - is a specification of levels that represents relationships between different attributes within a hierarchy
- Attribute hierarchies describe relationships among dimension attributes
  - E.g. products fall within brands; brands within categories
    - *these rules can be inferred without referring to actual data*
  - Such hierarchies are nice and easy to work with
- Another form of hierarchy may exist among instances of dimensions
  - E.g. employees report to other employees, companies own other companies
  - Finally, there may be any number of levels in the hierarchy









Dimensions

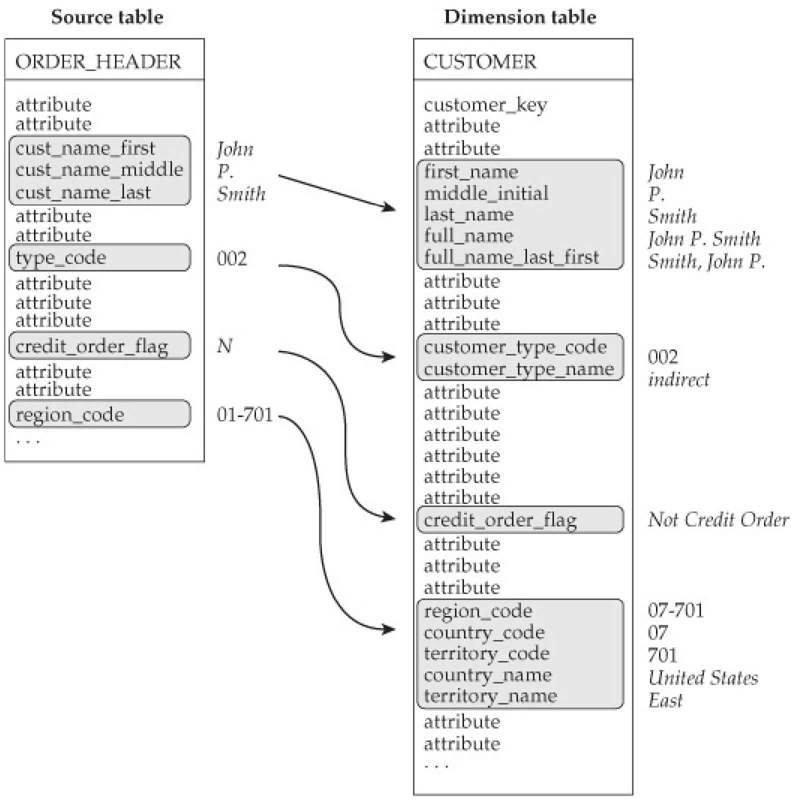
There are four major types of hierarchies:

- Balanced hierarchies
  - *the branches of the hierarchy all descend to the same level, with each parent of a member at the level immediately above the member.*
- Unbalanced hierarchies
  - *branches of the hierarchy descend to different levels*
    - branches can also have inconsistent depths.
  - *in general, the concept of levels does not apply*
- Ragged hierarchies
  - *the logical parent member of at least one member is not in the level immediately above the member*
    - the parent of a member can come from any level above the level of the member, not just from the level immediately above.
    - the branches of the hierarchies can descend to different levels - also be referred to as ragged-balanced, because levels still exist
- Parent-child hierarchies
  - *A parent attribute describes a self-referencing relationship, or self-join, within a dimension main table*
    - are constructed from a single parent attribute.
  - *Only one level is assigned to a parent-child hierarchy, however depth can be investigated*

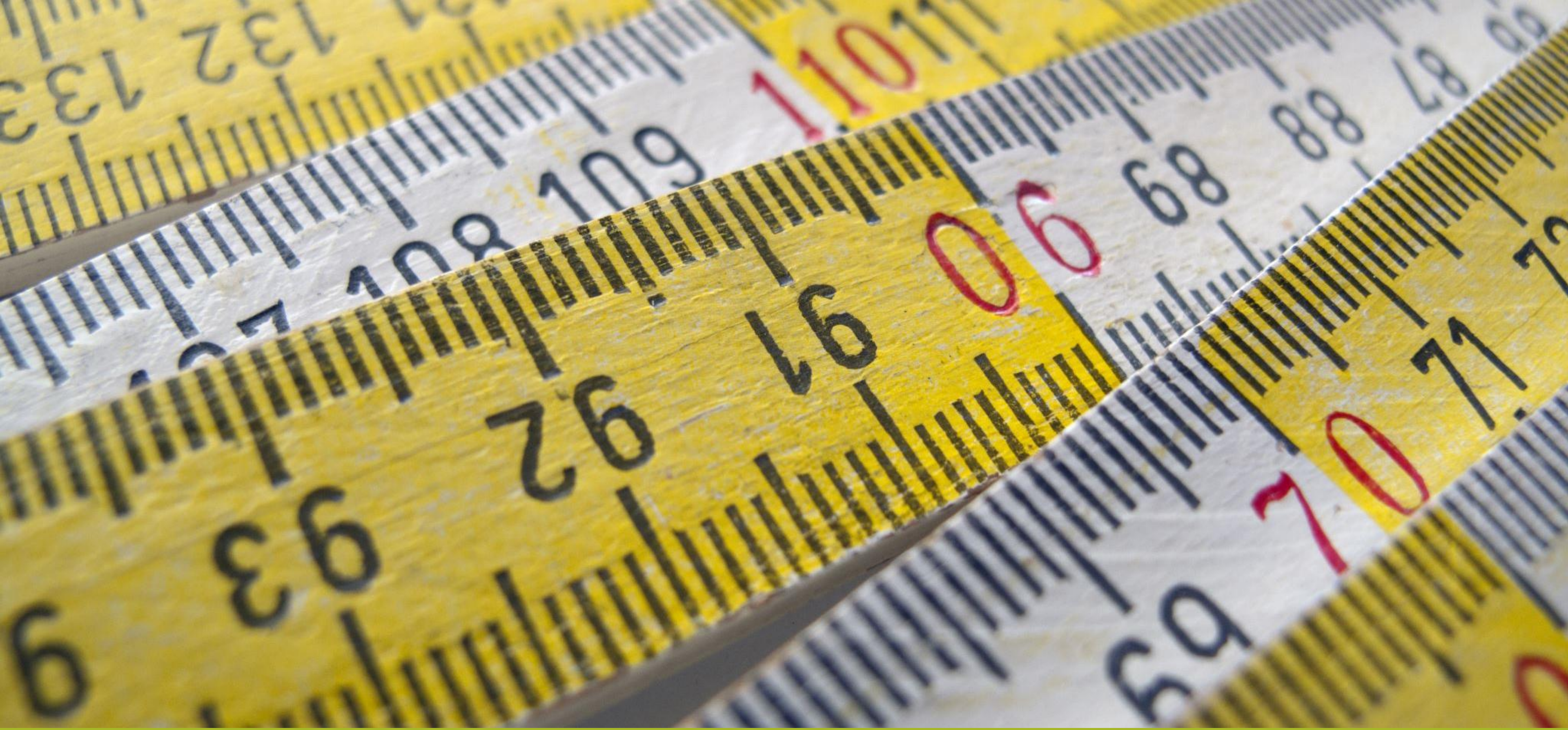
Dimensions

- Dimensions
  - are used for
    - *Selection of data*
    - *Grouping of data at the right level of detail*
  - consist of dimension values
    - *Product dimension have values ”milk”, ”cream”, ...*
    - *Time dimension have values ”1/1/2001”, ”2/1/2001”,...*
  - values may have an ordering
    - *Used for comparing cube data across values*
    - *Example: ”percent sales increase compared with last month”*
    - *Especially used for Time dimension*
- Dimensions are sometimes called the “soul” of the data warehouse
  - they contain the entry points and descriptive labels that enable the DW/BI system to be leveraged for business analysis

Dimensions



Source table
ORDER_HEADER
attribute
attribute
cust_name_first
cust_name_middle
cust_name_last
attribute
attribute
type_code
attribute
attribute
attribute
credit_order_flag
attribute
attribute
region_code
. . .
John
P.
Smith
002
N
01-701
Dimension table
CUSTOMER
customer_key
attribute
attribute
first_name
middle_initial
last_name
full_name
full_name_last_first
attribute
attribute
attribute
customer_type_code
customer_type_name
attribute
attribute
attribute
attribute
attribute
attribute
credit_order_flag
attribute
attribute
attribute
region_code
country_code
territory_code
country_name
territory_name
attribute
attribute
. . .
John
P.
Smith
John P. Smith
Smith, John P.
002
indirect
Not Credit Order
07-701
07
701
United States
East



Measures and aggregates

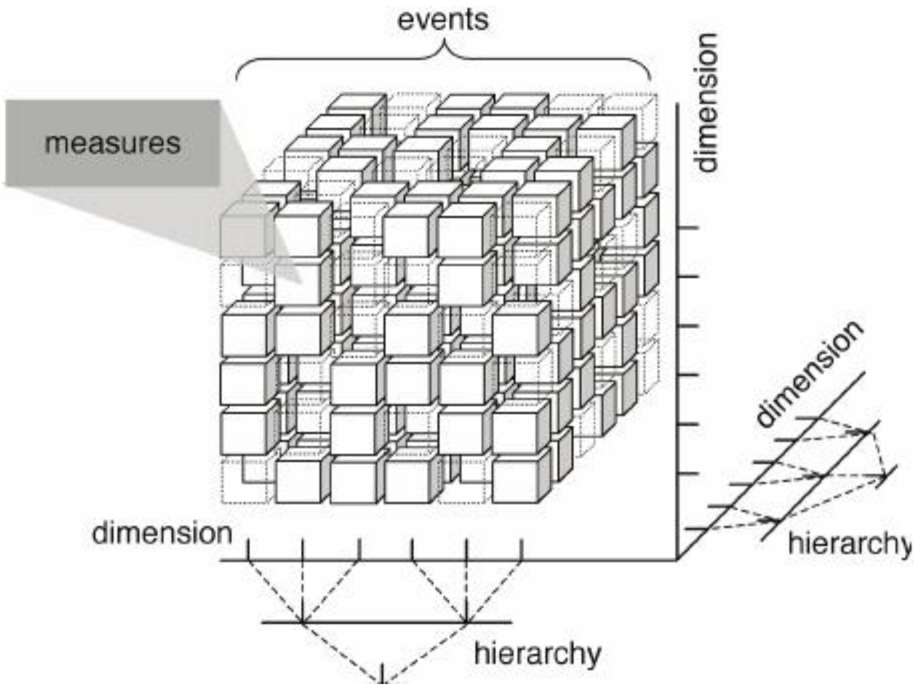
Facts

- Facts
  - A fact is an element of the multi-dimensional space
    - *Associates a set of dimension elements with measures*
  - „cube cell“
    - *Granularity relates to dimensions*
      - Fact is uniquely identified through a combination of dimension elements
    - *Granularity should be defined – depending on user requirements and data availability*
    - *Single granularity for a fact table*
  - Qualifying and quantifying information

Measures

- Measures
  - Quantifying information
    - *Usually numeric*
    - *in simple terms measures are normally the elements you want to add up in reporting*
  - Examples:
    - *sales figures*
    - *turnover*
    - *measurements (temperature, rainfall, ...)*

Multidimensional Data



Measures

Measure has two components

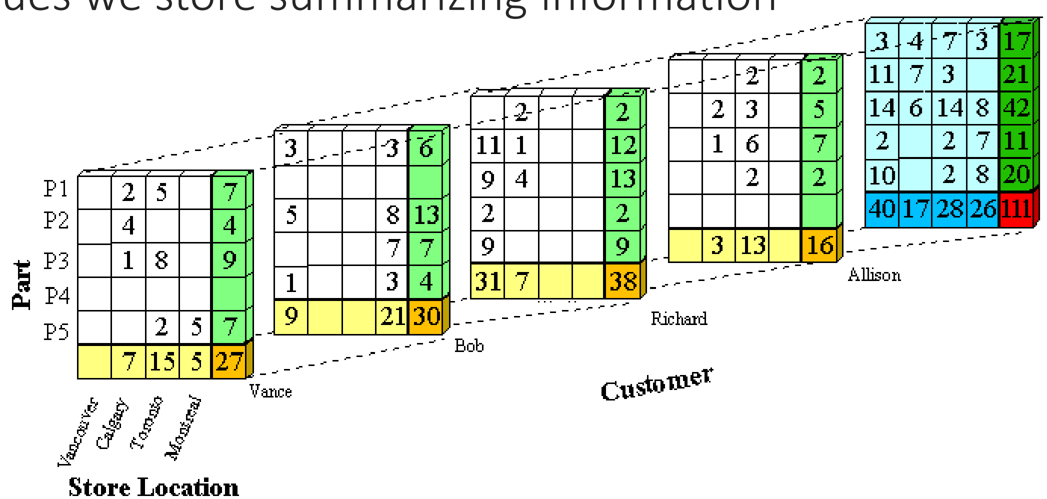
- Numerical value: (sales price)
- Aggregation formula (SUM): used for aggregating/combining a number of measure values into one

Measure value is determined by dimension value combination

- *Measure value should be meaningful for all aggregation levels*

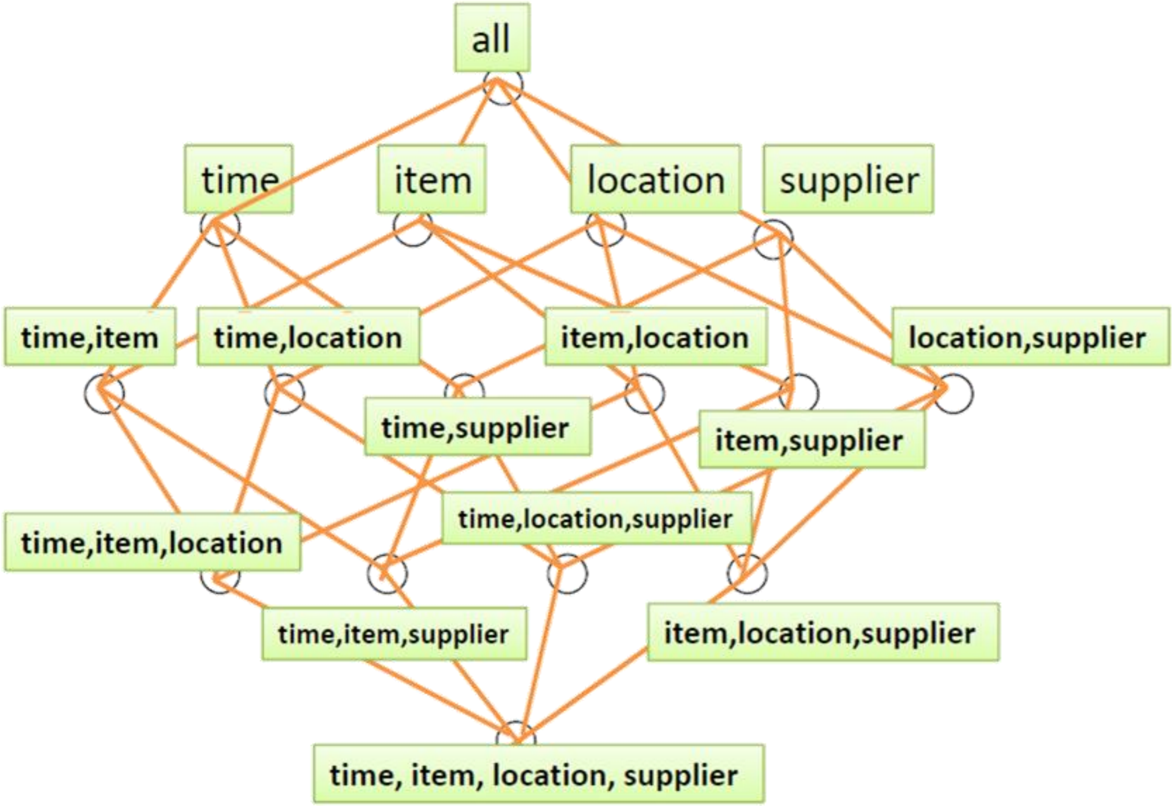
Aggregates

- together with measure values we store summarizing information



Multi-dimensional data

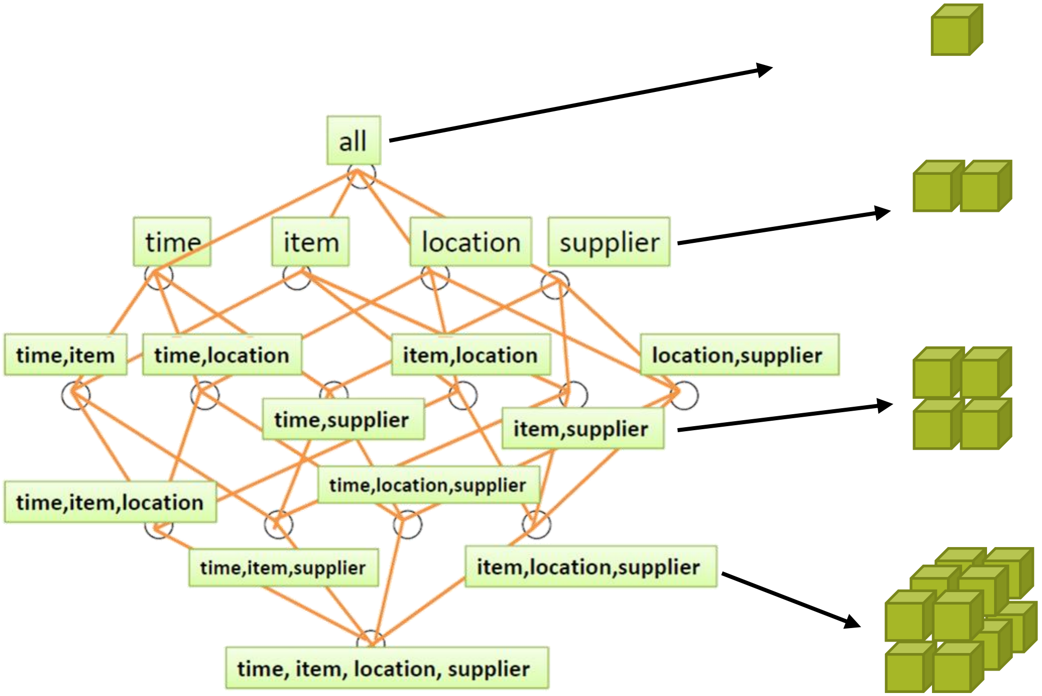
- Data cube
  - Lattice of cuboids



Multi-dimensional data

Data cube

- Lattice of cuboids



Multi-dimensional data

Lowest level of summarization is called the base cuboid

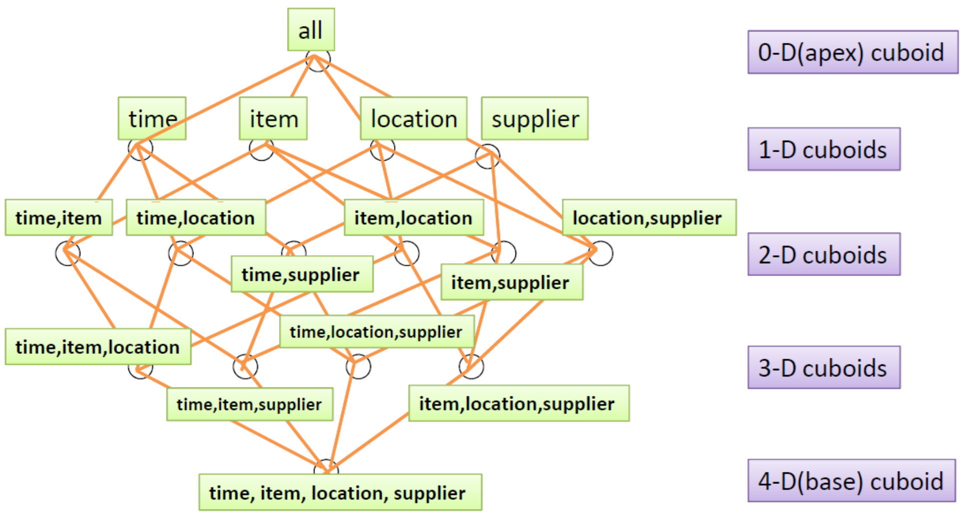
- In general N-D, where N is the number of dimensions
- E.g. base cuboid for the given *time*, *item*, *location*, and *supplier* dimensions - 4-D cuboid

M-D cuboid

- M dimensions, with N-M summaries
- E.g. 3-D cuboid for time, item, and location - summarized for all suppliers

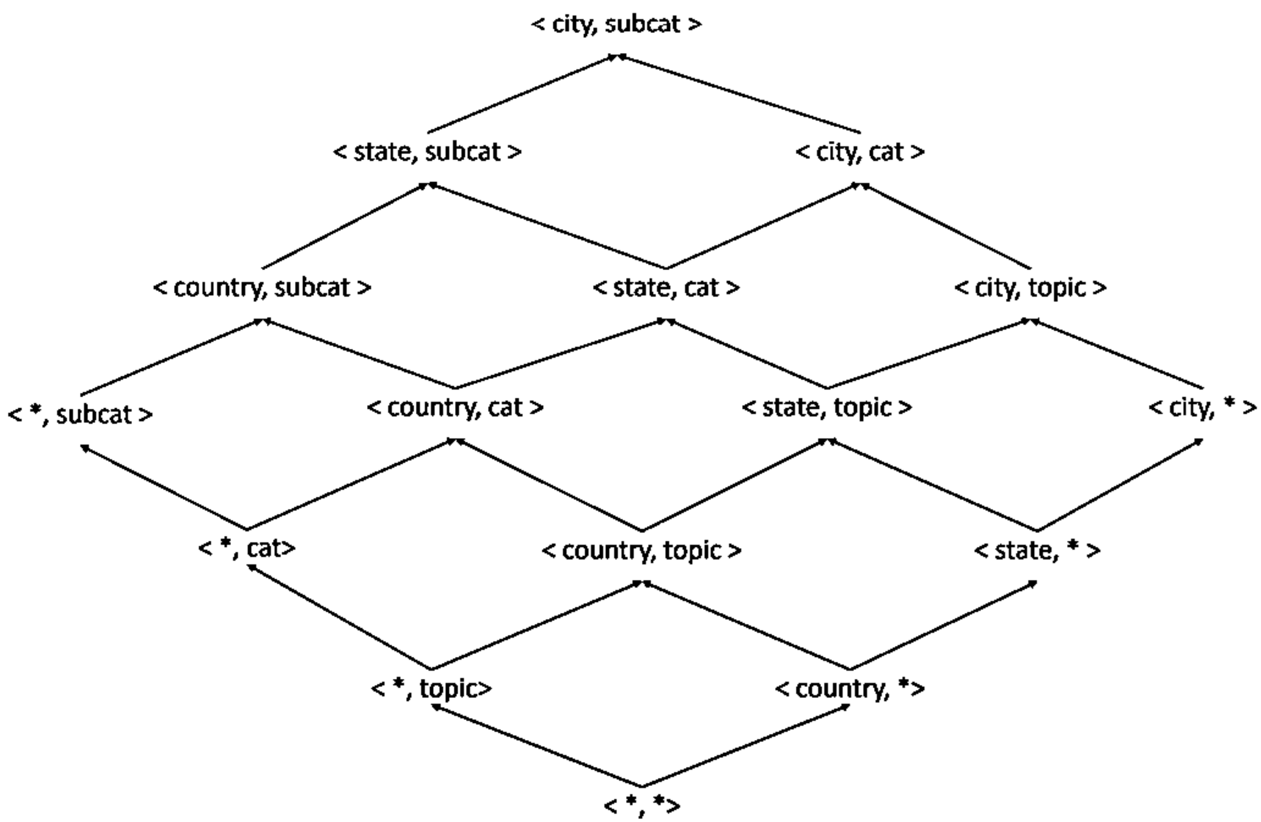
The 0-D cuboid

- holds the highest level of summarization
  - *called the apex cuboid*
- E.g. total sales summarized over all four dimensions
- typically denoted by all



Multi-dimensional data

- 2 dimensions
  - location
    - *country*
    - *state*
    - *city*
  - topic
    - *topic*
    - *cat*
    - *subcat*

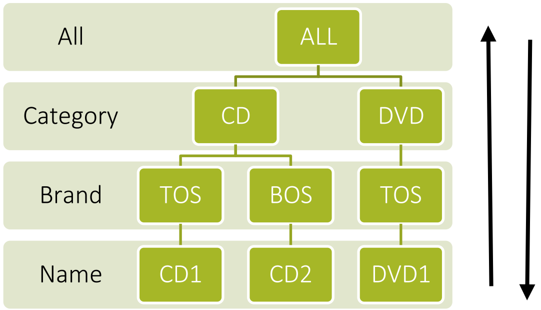


Multi-dimensional data

GROUP BY

- <>
- <Name>
- <Brand>
- <Category>
- <Name, Category>
- <Name, Brand>
- <Brand, Category>
- <Name, Brand, Category>

Navigation paths



Measures

- Measure types
  - Additive
    - *measures that can be added across any dimension.*
    - *For example total sales amount can be added across all dimensions*
  - Non-additive
    - *measures that cannot be added across any dimension.*
    - *For example product's margin*
  - Semi-additive
    - *measures that can be added across some dimensions.*
    - *For example the bank account balance is simply a snapshot in time and cannot be added over time. However you could add multiple accounts of the same customer to get the total balance for that customer.*

Measures

Measures can be organized into three categories based on the kind of aggregate functions used:

- Distributive:
  - *An aggregate function is distributive if it can be computed in a distributed manner*
- Algebraic:
  - *Like distributive aggregate functions, algebraic functions can be evaluated in different stages, but requires a consolidating function at the final stage to accumulate the result generated in the previous stages.*
- Holistic:
  - *An aggregate function is holistic if there is no constant bound on the storage size needed to describe a subaggregate.*
    - There does not exist an algebraic function with M arguments (where M is a constant) that characterizes the computation.

Measures

Distributive measure

The result derived by applying the function to n aggregate values is the same as that derived by applying the function on all the data without partitioning

A function F() is distributive if there is a scalar function G() such that

$$F(\{X_{i,j}\}) = G(\{F(\{X_{i,j}|i = 1, \ldots, I\})|j = 1, \ldots, J\})$$

Algebraic measure

Can be computed by an algebraic function with M arguments (where M is a bounded integer), each of which is obtained by applying a distributive aggregate function

A function F() is algebraic if there is a m-valued function G()≠F() and m-valued function H()≠F() such that

$$F(\{X_{i,j}\}) = H(\{G(\{X_{i,j}|i = 1, \ldots, I\})|j = 1, \ldots, J\})$$

G() computes aggregates for each group

H() consolidates results for each group

Holistic

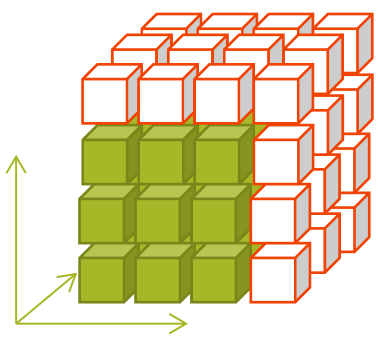
There is no constant bound on the storage size needed to describe a subaggregate

Multidimensional data

Multidimensional data

Aggregates

together with measure values we store summarizing information

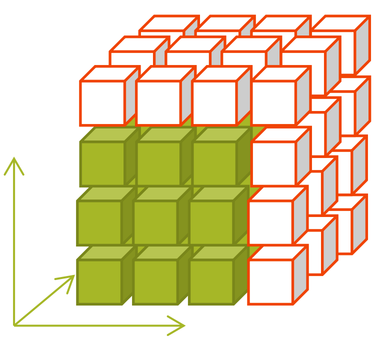


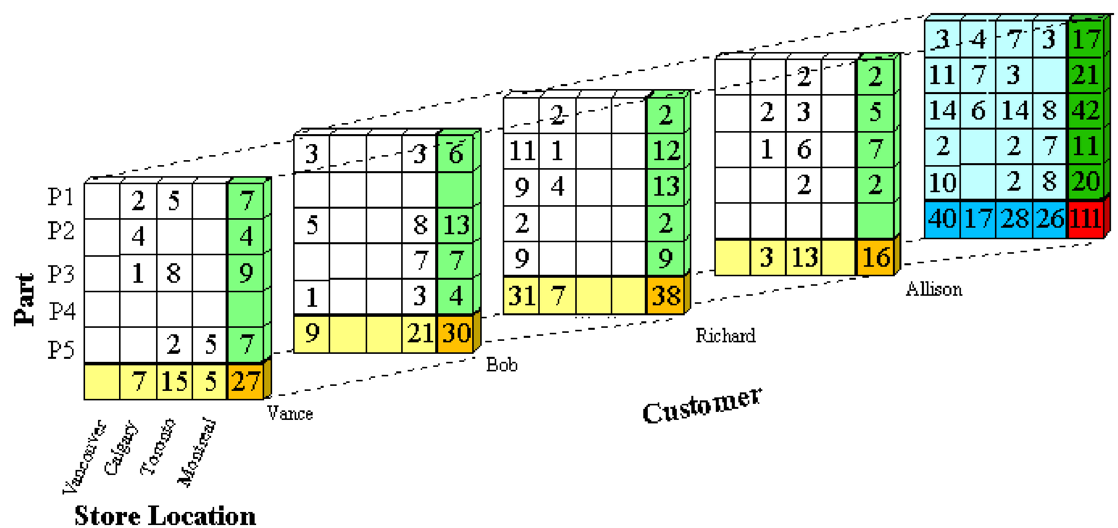
Location: Vancouver					
Time (quarters)	Items				
	TV	Computer	Phone	Security	Sum
Q1	605	825	14	400	1844
Q2	680	952	31	512	2175
Q3	812	1023	30	501	2366
Q4	927	1038	38	580	2583
Sum	3024	3838	113	1993	8968

Multidimensional data

Multidimensional data

- Aggregates
  - *together with measure values we store summarizing information*



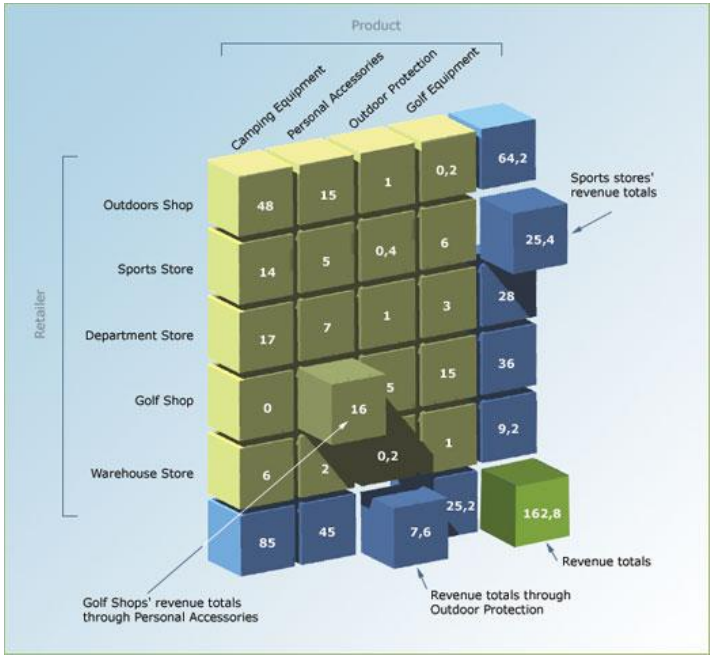


Data as a cube

Aggregations

OLAP queries also require the ability to aggregate data over different dimensions

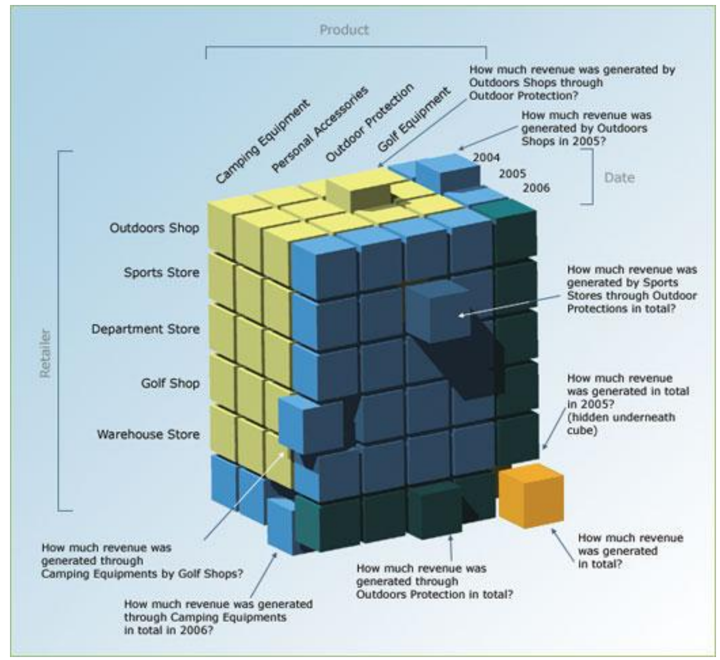
We need to have totals



data as a cube

Aggregations

- Many dimensions = many aggregates
  - *We need to calculate many different totals*
- What if some of these aggregates were precalculate during load?
  - *Storage is cheap, calculation is more expensive and time consuming*



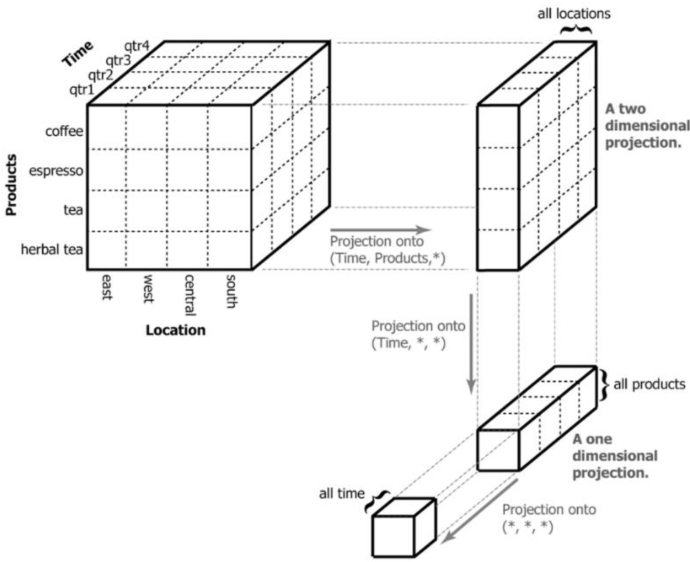
Data as a cube

Left to right

- How fine-grained data turns into a cube
  - *Calculation*
  - *First over locations, then over products, then over time*

Right to left

- How the cube containing all the data in one value is broken down
  - *In general, aggregations are irreversible*
    - Having the sum you don't know parts
  - *first by time, then by product, then by location*



Data as a cube

How you do you ask questions of such a dataset?

- You specify how you want it broken down and how to aggregate the measures.

Example

- Dataset of incomes for many different job titles, ages, genders, education levels, etc.
- Ask about the average income for men vs. women

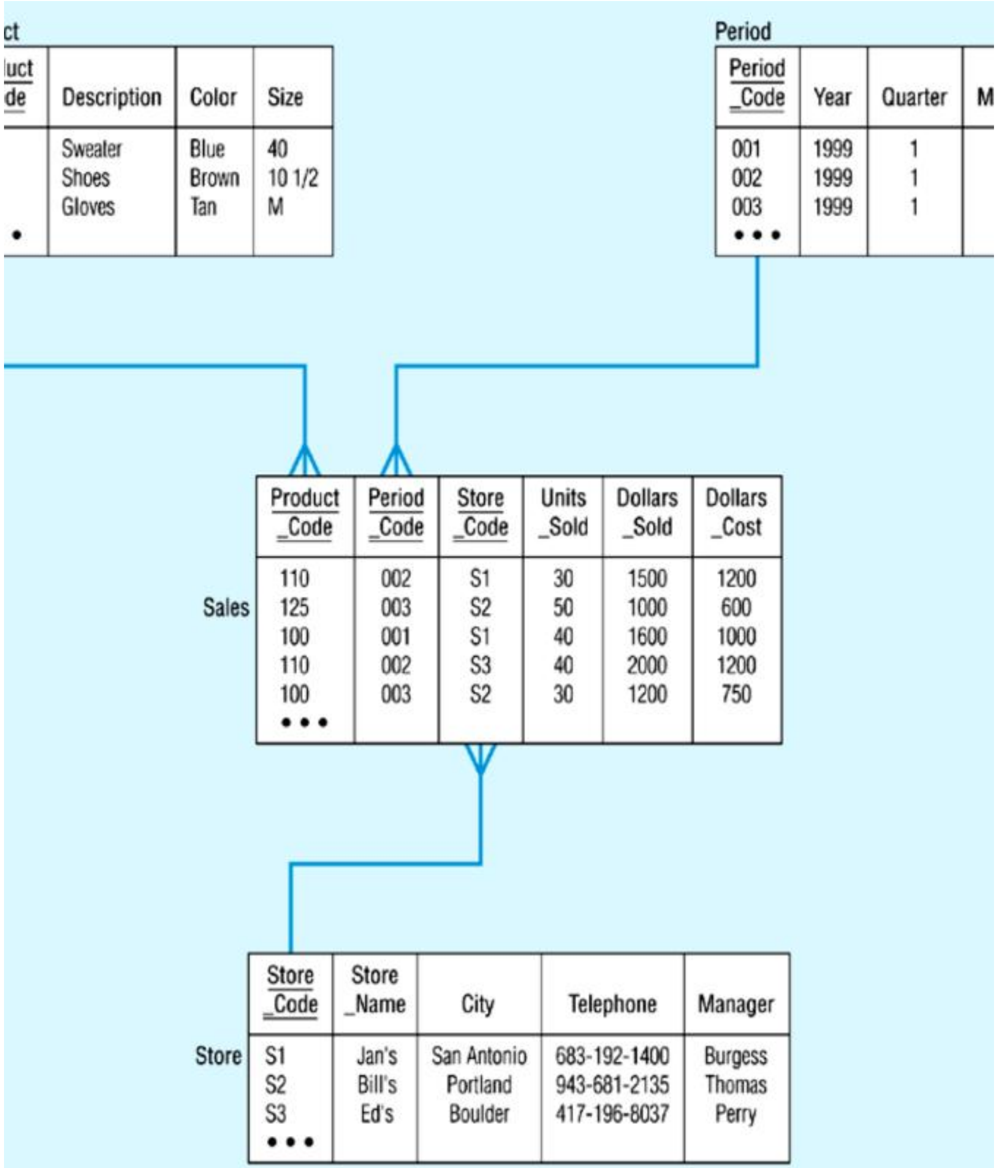
It's the dataset perspective

Process:

- Split the dataset along the gender dimension – into two groups.
- Calculate the averages for each gender.

Data as a cube

- This way, you can quickly ask many different questions:
  - What’s the average by education level?
  - How about gender and education level?
  - What about age?
  - Etc.
- Different aggregations in each slice are possible
  - average, median, minimum, maximum, etc.



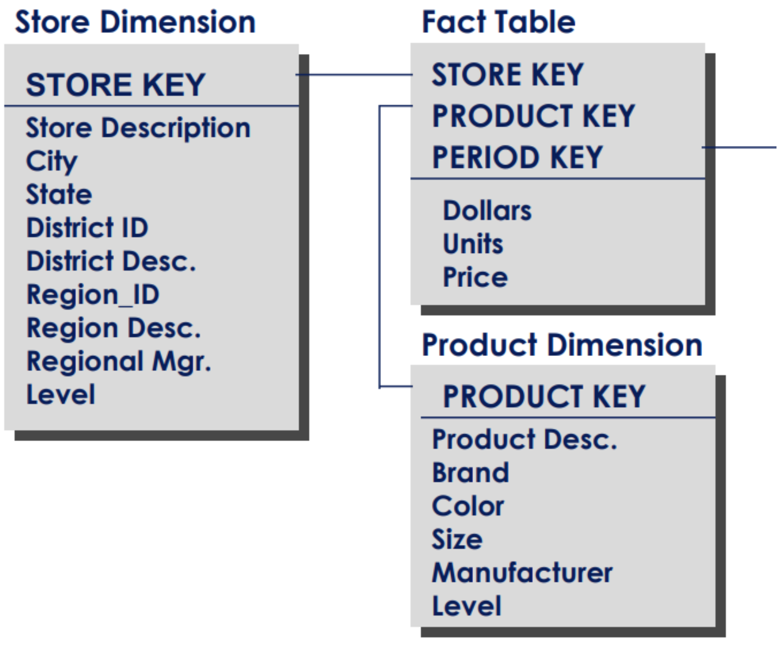
DW Schemas and Aggregates

- What about the aggregates?
  - E.g. for City, Manager

Star Schema

Star schema

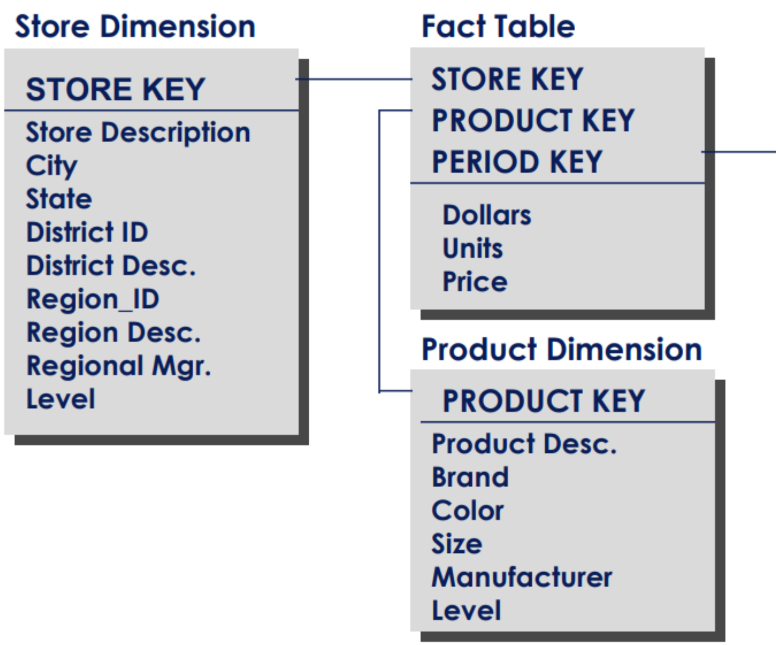
- A single fact table, with detail and summary data
- Fact table primary key has only one key column per dimension
  - *Each key is generated*
- Each dimension is a single table, highly denormalized



Star Schema

Level is needed whenever aggregates are stored with detail facts.

- Select
  - *A.dollars*
- from Fact_Table A where
  - *A.STORE_KEY in (select STORE_KEY from Store_Dimension B where region = "North" and Level = 2)*



Star Schema

Aggregates

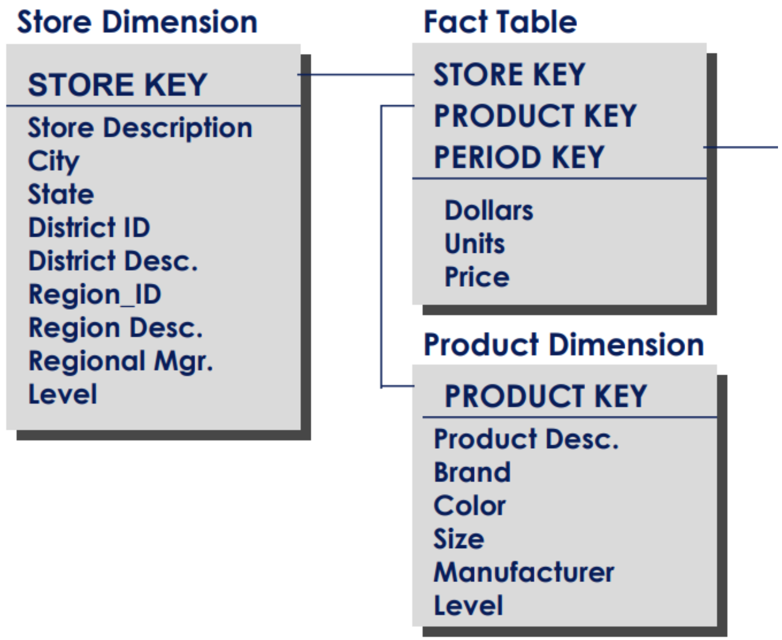
Advantages:

Easy to understand, easy to define hierarchies, reduces # of physical joins, low maintenance, very simple metadata

Disadvantages:

Summary data in the fact table yields poorer performance for summary levels, Huge dimension tables are a problem

Level is a problem – a potential for error.



Snowflake schema

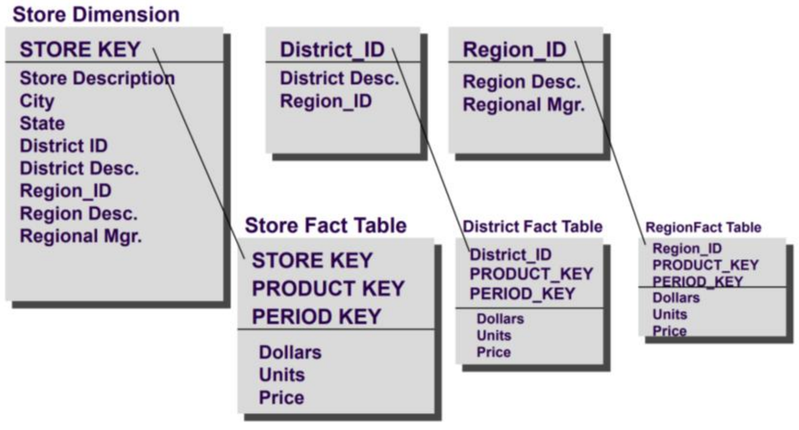
Snowflake aggregates

No LEVEL in dimension tables

Dimension tables are normalized by decomposing at the attribute level

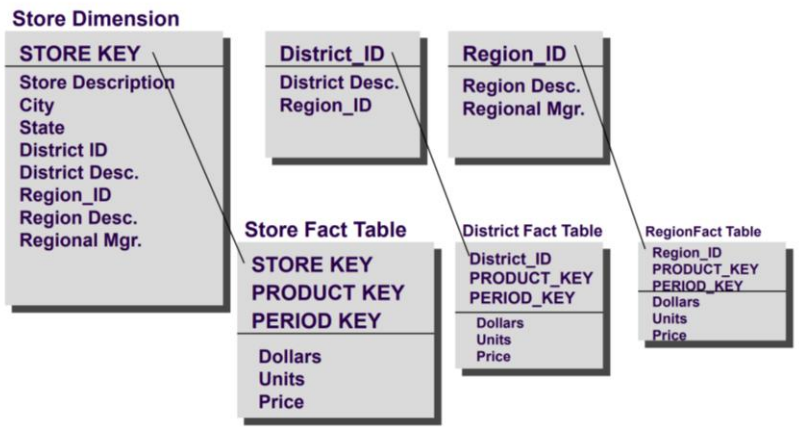
Each dimension table has one key for each level of the dimensionís hierarchy

The lowest level key joins the dimension table to both the fact table and the lower level attribute table



Snowflake schema

- Aggregates
  - Advantage:
    - *Best performance when queries involve aggregation*
  - Disadvantage:
    - *Complicated maintenance and metadata, explosion in the number of tables in the database*



Fact Constelation Schema

Constelation aggregates

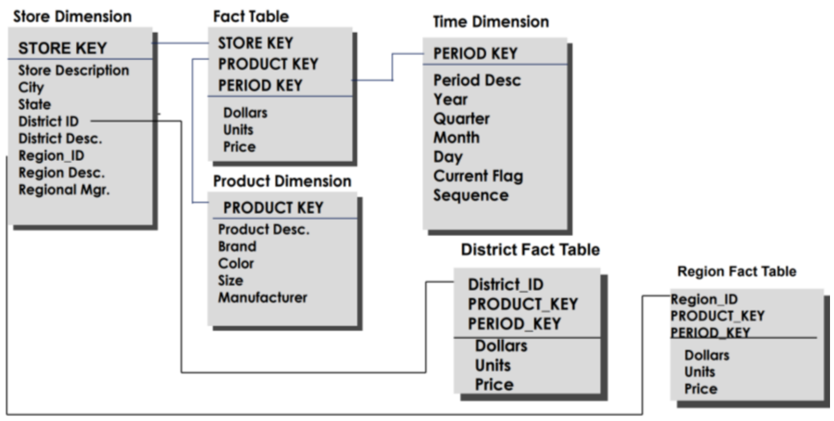
aggregate tables are created separately from the detail

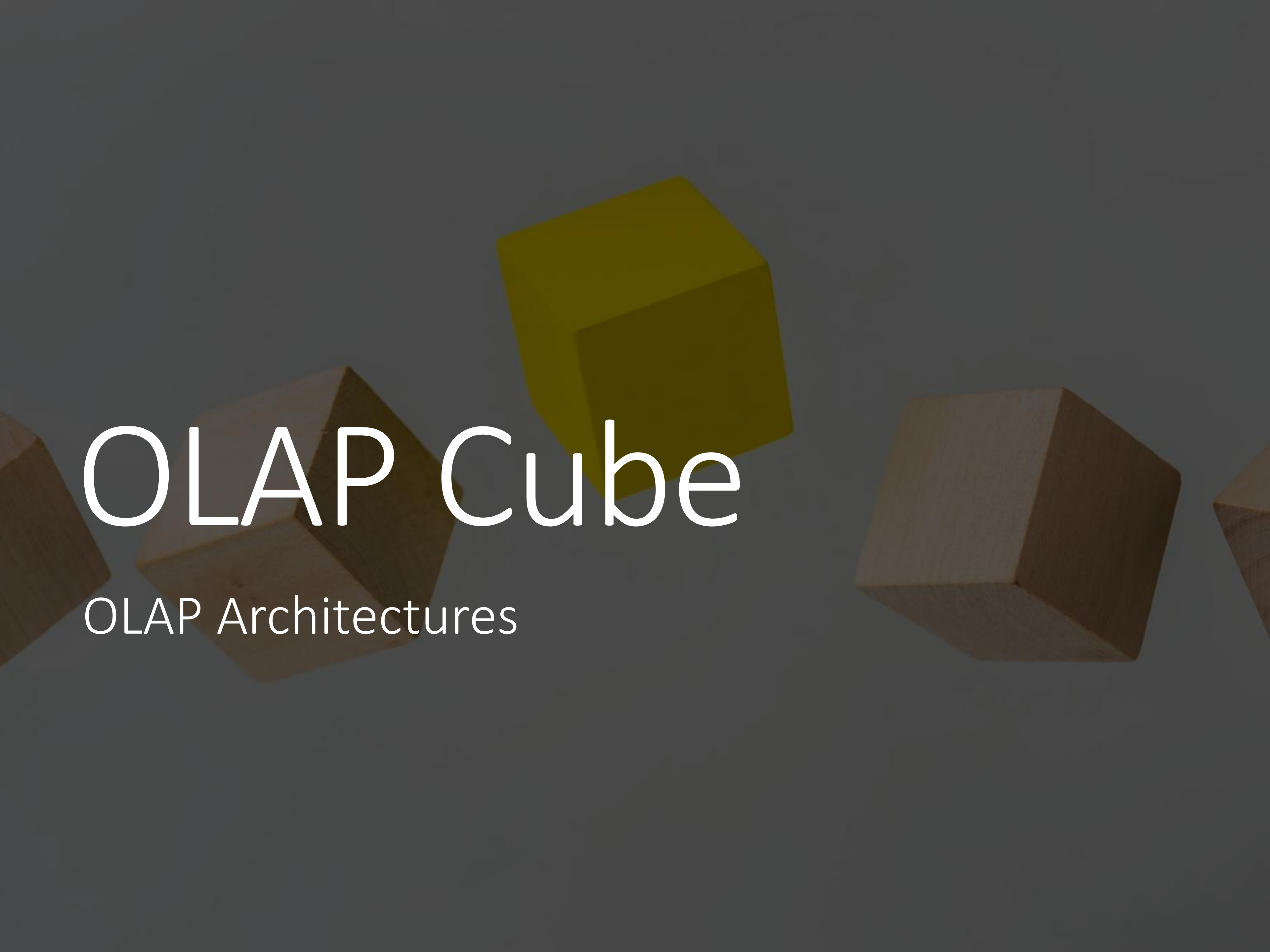
Advantage:

No need for the "Level" indicator in the dimension tables, since no aggregated data is stored with lower-level detail

Disadvantage:

Dimension tables are still very large in some cases, which can slow performance; front-end must be able to detect existence of aggregate facts, which requires more extensive metadata





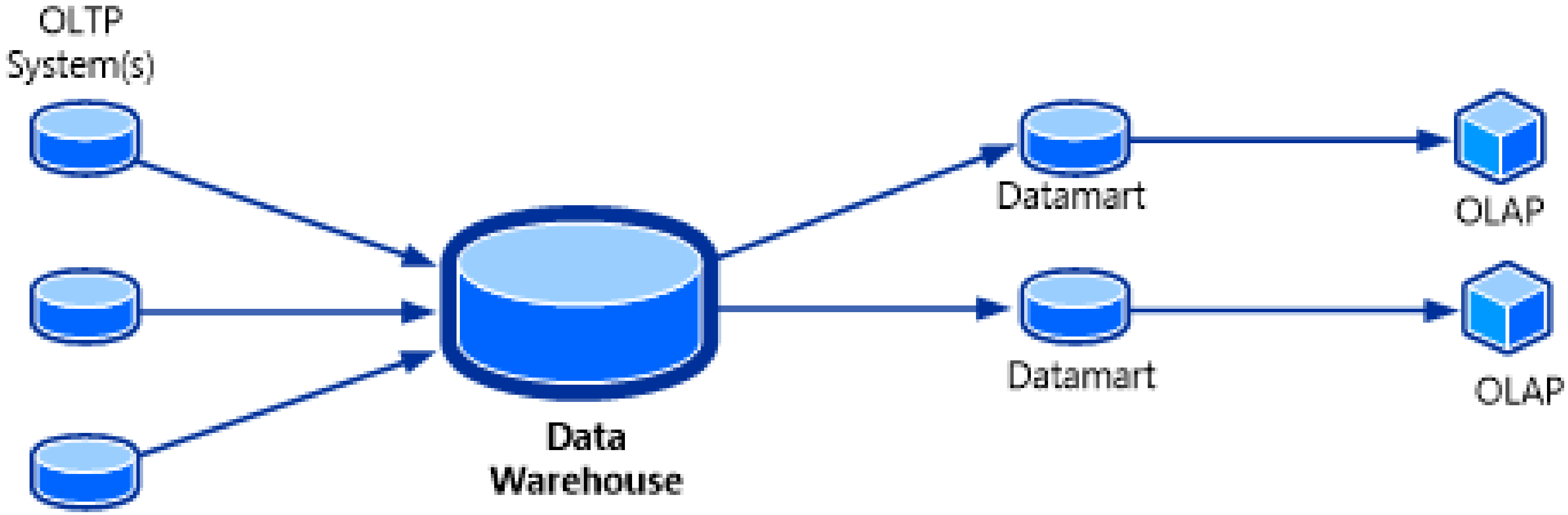
OLAP Cube

OLAP Architectures

OLAP cubes

multidimensional cube or hypercube

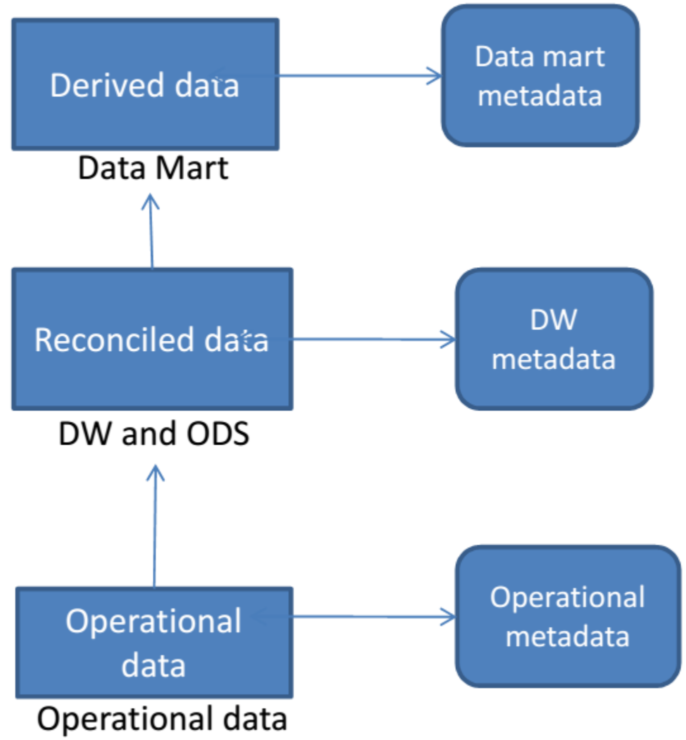
final piece of the puzzle for a data warehousing solution



Data and Metadata

Types of data in DW env.:

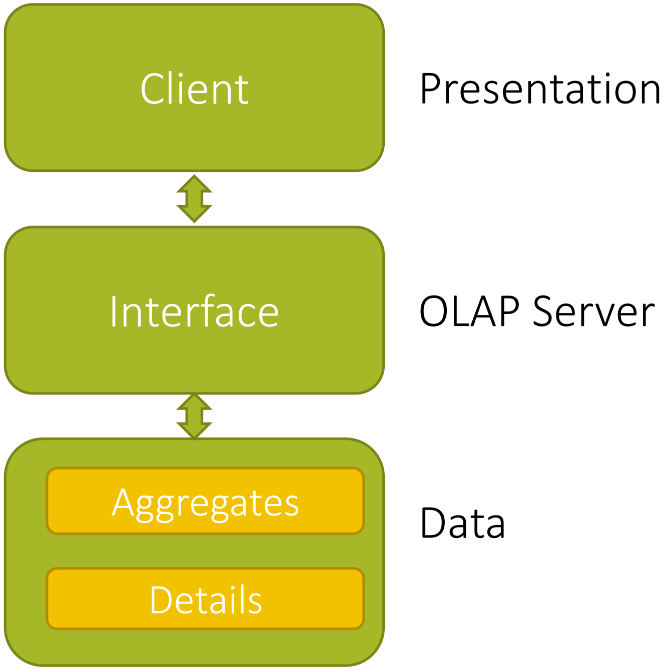
- Derived data
  - Data that had been selected, formatted, and aggregated for DSS support
- Reconciled data
  - Detailed, current data intended to be the single, authoritative source for all decision support
  - Ensured consistency
- Operational data
  - Source data



OLAP Architecture

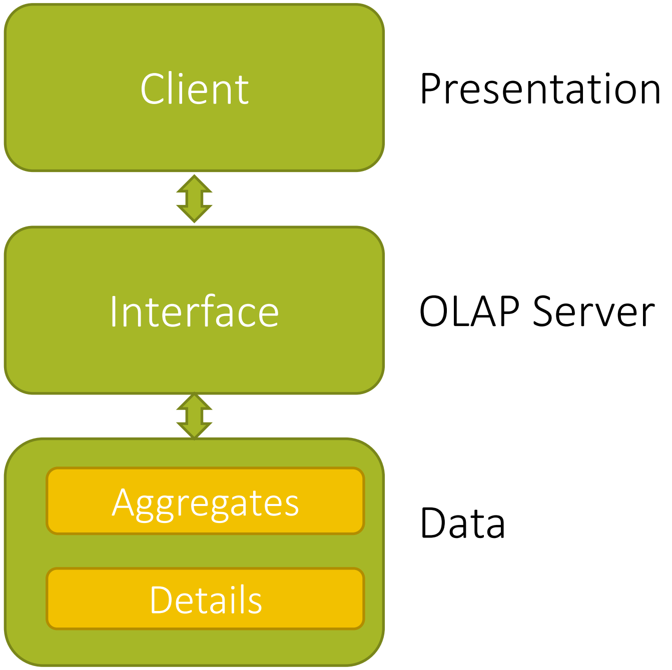
What kind of data we have?
- Detailed data
- Data summaries

What do we need to do with it?
- Storage
- Query
- Presentation



OLAP Architecture

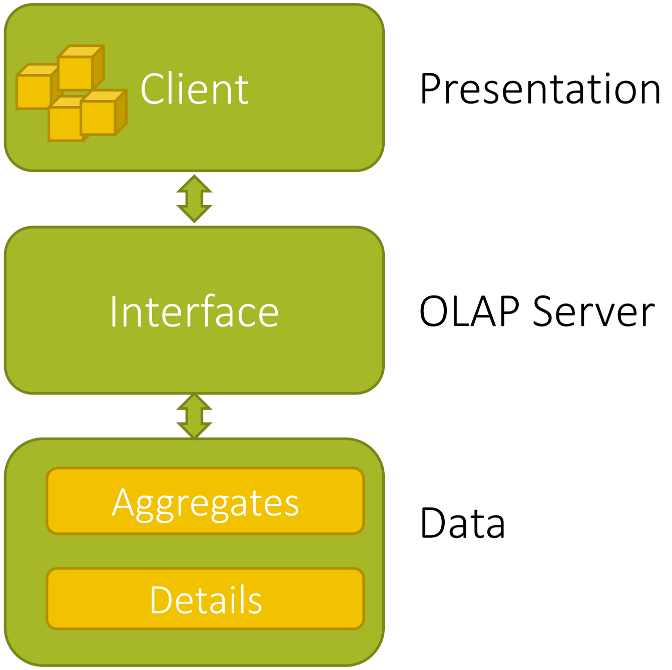
- How to store this data?
  - Detailed data
  - Summaries
- How to present this data?
  - Access



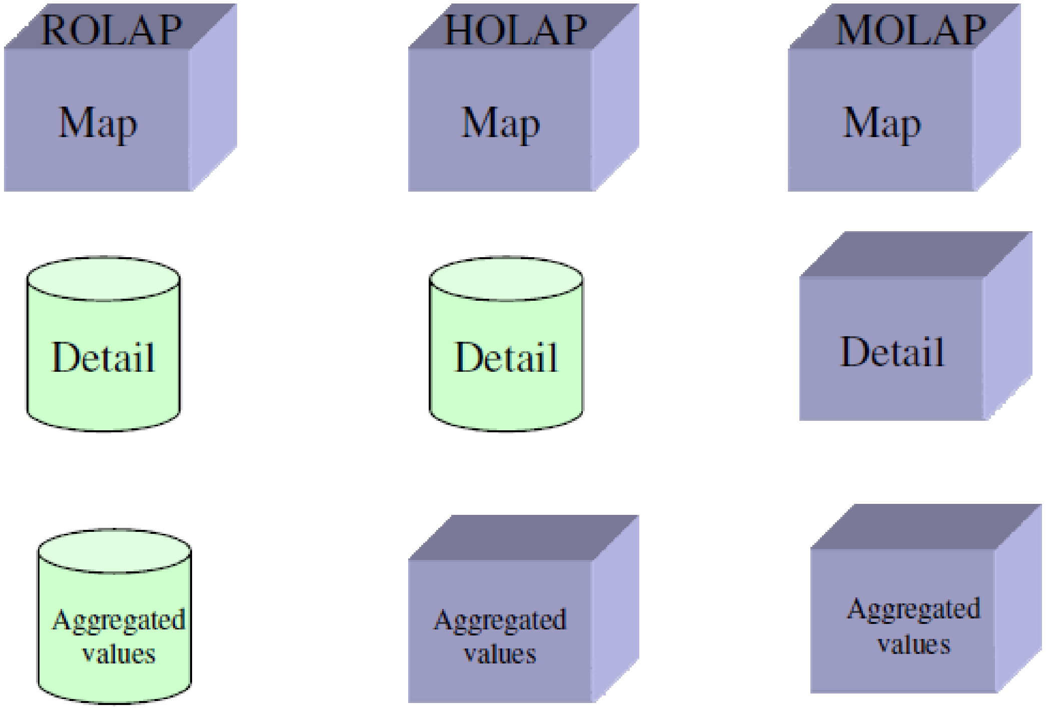
OLAP Architecture

Access

- Multidimensional perspective
  - *Representational gap*
- OLAP cubes
  - *OLAP operations*
  - *Roll-up, drill-down, etc.*



OLAP Architecture



MOLAP

MOLAP (Multidimensional OLAP)

- based on multi-dimensional database MDDB
- usually considered the OLAP standard

Data in arrays

- rather than storing information as records, and records in tables,
- MDDBs (logically) stores data in arrays
  - *each cell in the array is formed by the intersection of all dimensions.*

OLAP operation is multidimensional retrieval in data cube

MOLAP

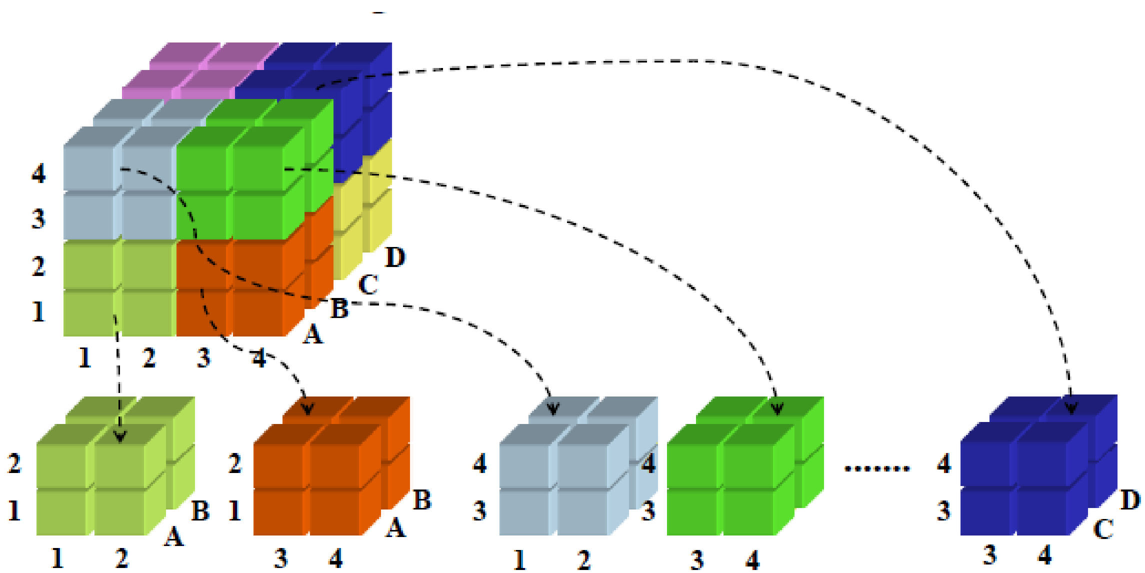
- Exemplar implementations
  - N-dim arrays
  - Hash table (SQL Server)
    - *Hash Table is a data structure which stores data in an associative manner*
      - data is stored in an array format, where each data value has its own unique index value
      - access to data is very fast if you know the index of the desired data
  - Quad tree
    - *Quadtrees are trees used to efficiently store data of points on a two-dimensional space.*
      - each node has at most four children.
  - K-D tree (K-Dimensional Tree)
    - *is a binary search tree where data in each node is a K-Dimensional point in space (space partitioning data structure for k-dimensional space)*
      - A non-leaf node in K-D tree divides the space into two parts, called as half-spaces
        - Points to the left are represented by the left subtree; points to the right of the space are represented by the right subtree

MOLAP

- MOLAP Servers
  - optimization techniques:
    - *Chunking*
    - *Two-level storage representation:*
      - dense cubes are identified and stored as array structures
      - sparse cubes employ compression techniques
    - *Materialized cubes*
    - *Partitions*

MOLAP

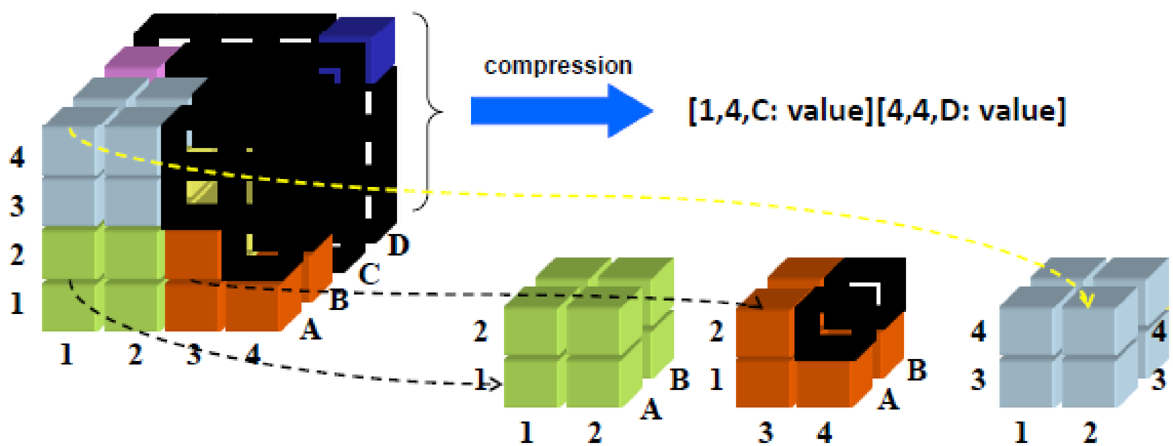
- Cube chunking
  - Enable parallel processing
  - Allow for better memory management



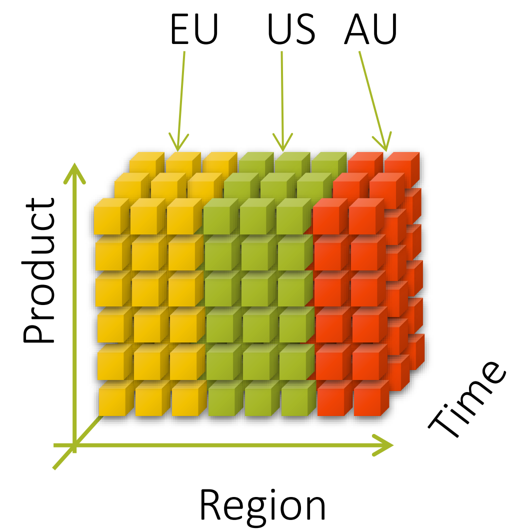
MOLAP

Compresion

- Store cells that contain some values – non empty



MOLAP



Partitioned Cubes

To overcome the space limitation of MOLAP, the cube can be partitioned

One logical cube of data can be spread across multiple physical cubes on separate (or same) servers

The divide&conquer cube partitioning approach helps alleviate the scalability limitations of MOLAP implementation

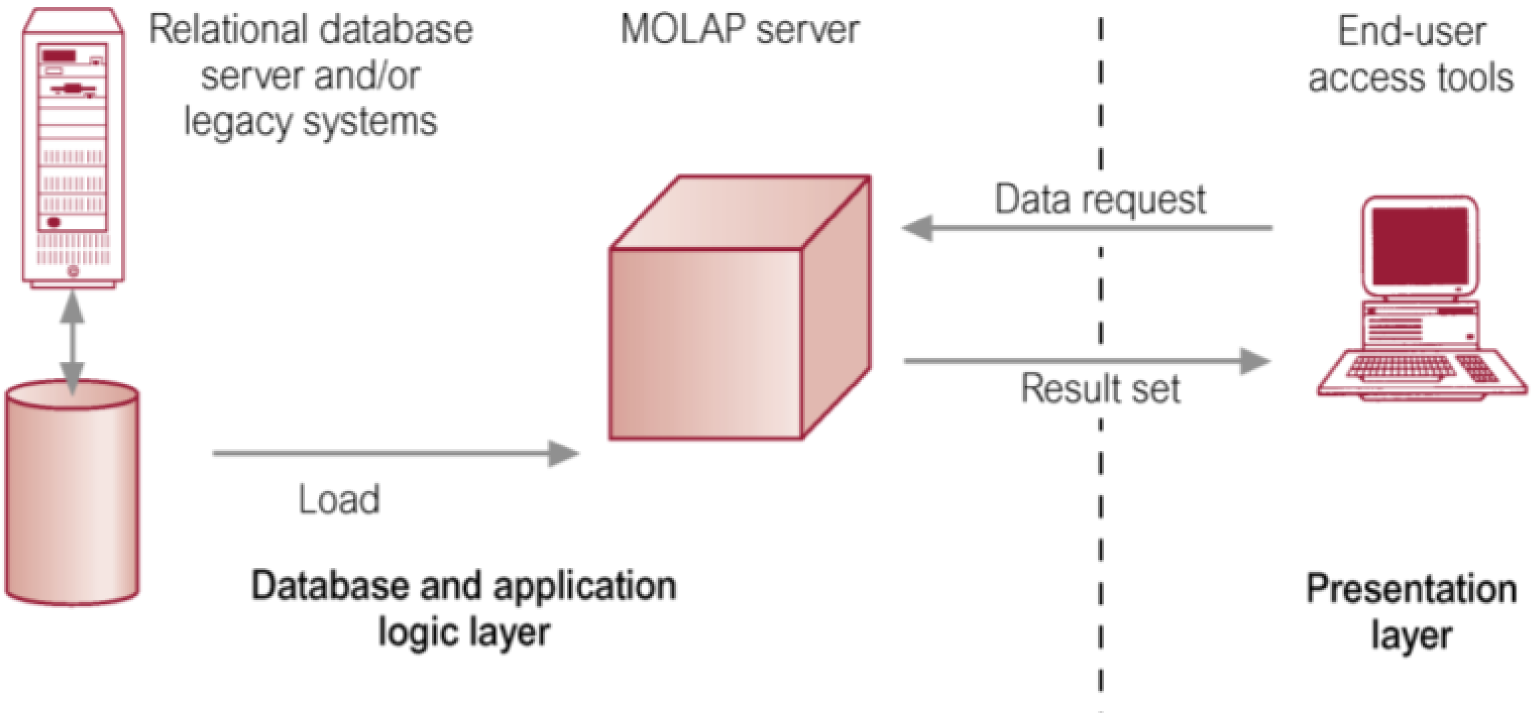
Advanced MOLAP implementations

Allow to distribute queries across partitioned cubes (potentially on distinct server platforms) for parallel retrieval

Ideal cube partitioning is completely invisible to end users

Performance degradation does occur in case of a join across partitioned cubes

MOLAP


Relational database server and/or legacy systems
MOLAP server
End-user access tools
Data request
Result set
Load
Database and application logic layer
Presentation layer

MOLAP

Advantages ?

Disadvantages ?

Main Aspects:

- Access
- Storage
- Loading
- Flexibility

MOLAP

Advantages of MOLAP Servers:

- Main advantage is the speed of query processing
  - *Multi-dimensional views are directly mapped to data cube array structures for efficient access to data*
- Can easily store subaggregates
- All calculations are pre-generated when the cube is created

Disadvantages of MOLAP Servers:

- The major disadvantage of multidimensional model is the high overhead of storage and precalculating.
  - *Multidimensional array model is not efficient for the sparse data*
  - *Scalability problem in the case of larger number of dimensions*
    - Enormous amount of overhead
      - An input file of 200 MB can expand to 5 GB with calculations
- Requires additional investment
  - *Cube technology are often proprietary*
- Further, when the dimensions are changed, the big data cube needs to be reconstructed which is not adaptive for real-time OLAP.

MOLAP

- Disadvantages
  - Scalability
    - *MDDB are good for handling summary data but are not always suited to handling large amounts of detailed data.*
      - MDDB are well suited for access to data typically less than 10-20 GB, involving a small number of dimensions (typically less than about 10), and where data access are well defined.
    - *Often have difficulty scaling when the size of dimensions becomes large*
      - The breakpoint is typically around 64,000 cardinality of a dimension.

MOLAP

Example

- Logical model consists of four dimensions:
  - *customer, product, location, and day*
- In case of 100 000 customers, 10 000 products, 1 000 locations and 1000 days
  - *the data cube will contain 1 000 000 000 000 000 cells!*
- Huge number of cells can be empty:
  - *a customer is not able to buy all the products in all the locations*

MOLAP

- Disadvantages
  - No consensus
    - *Unlike the relational model, it is not agreed-upon multi-dimensional database model*
  - No standard access method
    - *MDDBs have no standard access method (such as SQL) or APIs*

MOLAP

- MOLAP Implementations
  - No standard query language for querying MOLAP
    - *No SQL !*
    - *in a MOLAP environment*
      - there are no tables,
      - there are no traditional relational structures,
      - hence ANSI SQL can not be used.
      - there is no standard query language for querying a MOLAP cube

  - Vendors provide proprietary languages allowing business users to create queries that involve pivots, drilling down, or rolling up.
    - *e.g. MDX of Microsoft*

MOLAP

Disadvantages

Long pre-calculation time

Pre-calculating (e.q. summarizing)

all derived values will result in very fast retrieval times and that data explosion (large amounts of new data) does not matter since disk space is relatively cheap

may make MOLAP inappropriate for large or distributed applications with more than five dimensions because of long pre-calculating time.

Information structure modification

The modification of information structure is very difficult and almost always forces the total rebuilding of MDDB structure

MOLAP

Advantages

- Natural representation
  - *Natural representation of multi-dimensional structures used in OLAP and in consequence good efficiency of dimensional analyses*
- Short response time
  - *Short response time for queries with multi-dimensional calculations*
- MOLAP usage
  - *MOLAP can be used for department data marts*

Challenges in MOLAP

- Efficient storage and access
  - Proper storage for multidimensional OLAP data
  - Proper access
- Storing large arrays for efficient access
  - Storage organization, Chunking, Compressing sparse arrays
- Creating array data from data in tables
  - Efficient techniques for cube computation
- Handling new data
- Handling updates

ROLAP

Presentation layer

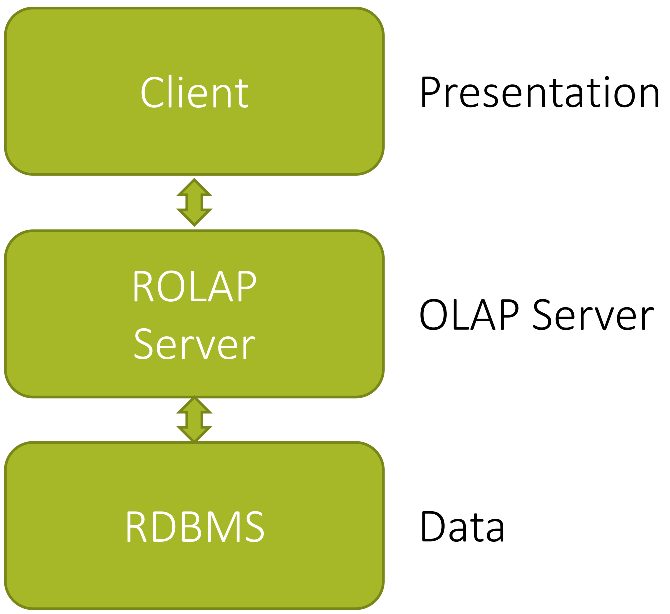
provides the multi-dimensional view

ROLAP Server generates

SQL queries, from the OLAP requests, to query the RDBMS

RDBMS

Data is stored in RDBs



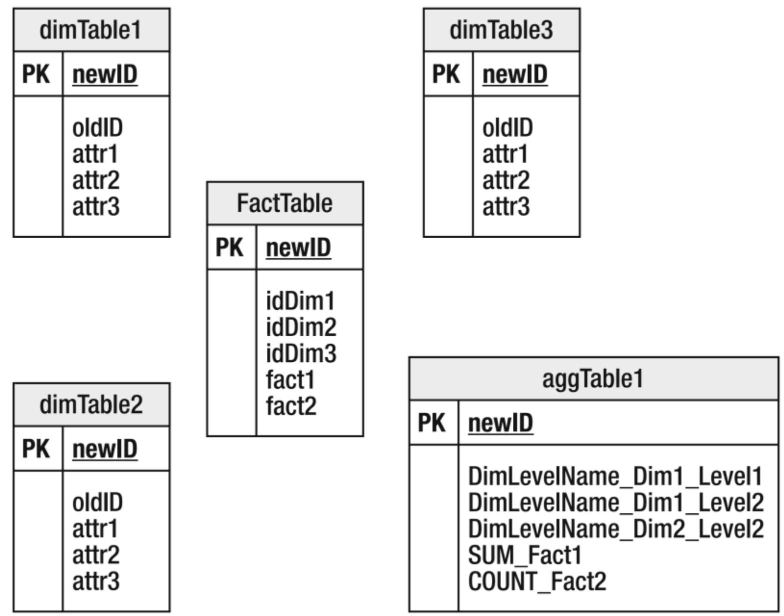
ROLAP

- Why ROLAP?
  - Issue of scalability
    - *Curse of dimensionality for MOLAP*
    - *Deployment of significantly large dimension tables as compared to MOLAP using secondary storage*
    - *Aggregate awareness*
      - allows using pre-built summary tables by some front-end tools
    - *Star schema designs usually used to facilitate ROLAP querying*

ROLAP

OLAP data is stored in a relational database

- e.g. using a star schema
- the fact table is a way of visualizing as an “un-rolled” cube
- so where is the cube?
  - *it’s a matter of perception*
  - *visualize the fact table as an elementary cube*



ROLAP

How to create a “Cube” in ROLAP

- Cube is a logical entity containing values of a certain fact
  - *at a certain aggregation level*
  - *at an intersection of a combination of dimensions.*
- The following table can be created using 3 queries

Month_ID

Product_ID	SUM (Sales_Amt)	M1	M2	M3	ALL
	P1				
	P2		Part-II		Part-I
	P3				
	Total		Part-III		

ROLAP

ROLAP model employs relational database as OLAP engine, and the multidimensional operation is transformed as SQL queries.

The data cube is materialized as aggregation tables, and the multidimensional retrieval is implemented as table scan.

For ad-hoc OLAP queries, the relational join performance dominates the ROLAP performance.

The essential difference between MOLAP and ROLAP model is multidimensional addressing mechanism.

- *MOLAP model is designed with double mapping between dimensions and fact data by organizing fact data with the multidimensional array (even for NULL cells)*
- *ROLAP model simplifies the dimensional addressing mechanism with primary key and foreign key constraints*

ROLAP

Problem with the simple approach

Number of required queries increases exponentially with the increase in number of dimensions

It’s wasteful to compute all queries

In the example, the first query can do most of the work of the other two queries

If we could save that result and aggregate over Month_Id and Product_Id, we could compute the other queries more efficiently

“Work smart not hard”

ROLAP

ROLAP and Space Requirement

If one is not careful, with the increase in number of dimensions, the number of summary tables gets very large

Consider the example discussed earlier with the following two dimensions on the fact table:

Time:Day, Week, Month, Quarter, Year, All Days

Product: Item, Sub -Category, Category, All Products

A naïve implementation will require all combinations of summary tables at each and every aggregation level

Smart tools will allow less detailed aggregates to be constructed from more detailed aggregates

ROLAP

ROLAP servers

use a relational or post-relational database management system to store and manage warehouse data

optimization techniques:

Denormalization,

Materialized views,

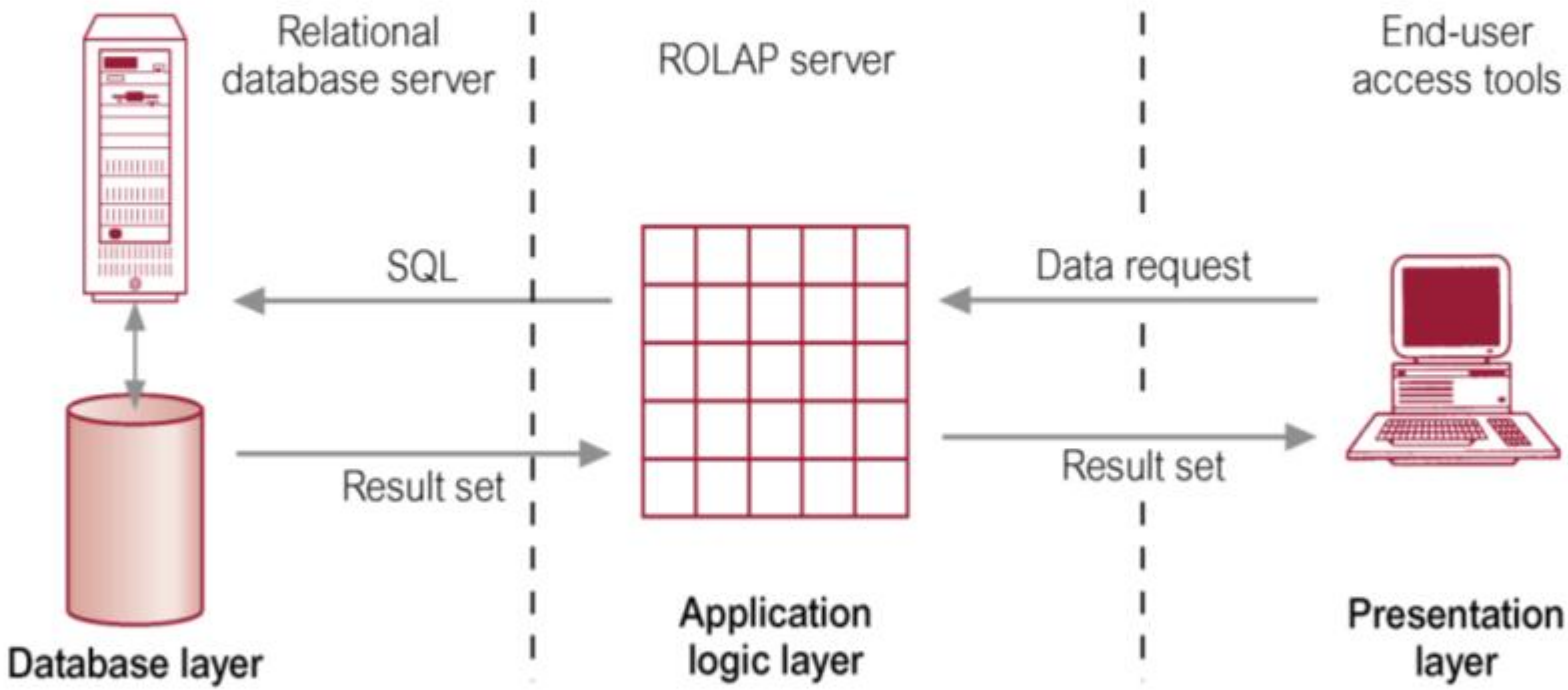
Partitioning,

Joins,

Indexes,

Query processing.

ROLAP



Relational database server
ROLAP server
End-user access tools
SQL
Data request
Result set
Result set
Database layer
Application logic layer
Presentation layer

ROLAP

Advantages ?

Disadvantages ?

Main Aspects:

- Access
- Storage
- Loading
- Flexibility

ROLAP

Advantages of ROLAP Servers:

- Scalable with respect to the number of dimensions,
- Scalable with respect to the size of data,
- Sparsity is not a problem (fact tables contain only facts),
- Mature and well-developed technology.

Disadvantage of ROLAP Servers:

- Worse performance than MOLAP,
- Additional data structures and optimization techniques used to improve the performance.

ROLAP

ROLAP Issues

- Maintenance.
- Non-standard hierarchy of dimensions.
- Explosion of storage space requirement.
- Aggregation pit-falls.

ROLAP

ROLAP Issue:

- Maintenance
  - *Summary tables are mostly a maintenance issue (similar to MOLAP) than a storage issue.*
  - *Notice that summary tables get much smaller as dimensions get less detailed (e.g., year vs. day).*
  - *Should plan for twice the size of the un-summarized data for ROLAP summaries in most environments.*
  - *Assuming "to-date" summaries, every detail record that is received into warehouse must aggregate into EVERY summary table.*

ROLAP

ROLAP Issue:

- Hierarchies
  - *Dimensions are NOT always simple hierarchies*
    - Dimensions can be more than simple hierarchies
      - i.e. item, subcategory, category, etc.
  - *The product dimension might also branch off by trade style that cross simple hierarchy boundaries such as:*
    - Looking at sales of air conditioners that cross manufacturer boundaries, such as COY1, COY2, COY3 etc.
    - Looking at sales of all “green colored” items that even cross product categories (washing machine, refrigerator, split- AC, etc.).
    - Looking at a combination of both.

ROLAP

- ROLAP Issue:
  - Storage Space Explosion
    - *Summary tables required for non-standard grouping*
    - *Summary tables required along different definitions of year, week etc.*
    - *Brute force approach would quickly overwhelm the system storage capacity due to a combinatorial explosion.*

ROLAP

How to reduce summary tables?

Many ROLAP products have developed means to reduce the number of summary tables by:

Building summaries on-the-fly as required by end-user applications.

Enhancing performance on common queries at coarser granularities.

Providing smart tools to assist DBAs in selecting the "best” aggregations to build i.e. trade -off between speed and space.

Materializing cubes

Materialization of Cuboids in ROLAP

Sorting, hashing, and grouping operations are applied to the dimension attributes in order to reorder and cluster related tuples,

Grouping is performed on some sub-aggregates as a “partial grouping step“

these partial groupings may be used to speed up the computation of other sub-aggregates,

Aggregates may be computed from previously computed aggregates, rather than from the base fact table.

OLAP needs		MOLAP	ROLAP
User Benefits	Multidimensional View	√	√
	Excellent Performance	√	-
	Analytical Flexibility	√	-
	Real-Time Data Access	-	√
	High Data Capacity	-	√
MIS Benefits	Easy Development	√	-
	Low Structure Maintenance	-	√
	Low Aggregate Maintenance	√	-

R/M OLAP

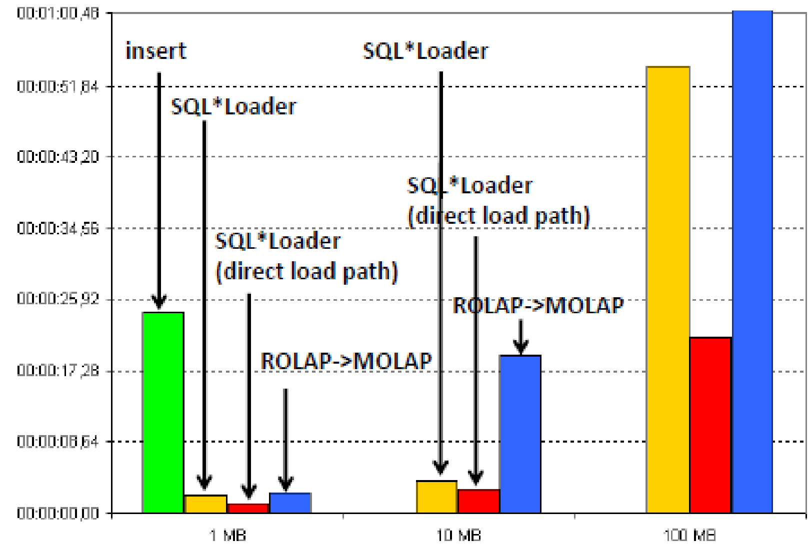
	MOLAP	ROLAP
Aggregation level	Require high level of aggregation otherwise are less useful	May be used at every level of aggregation. Aggregation level depends on analyses requirements
Number of dimensions	Bounded dimensionality practically up to 10 dimensions	Fluid dimensionality
Atomic data	Up to 10 - 20 GB	Even 100 times bigger than 10-20 GB
Response time for queries	Short time	Long time. The longer, the more complex is the structure and the number of data and indexes is bigger
Application	Data marts	Data warehouses

R/M OLAP

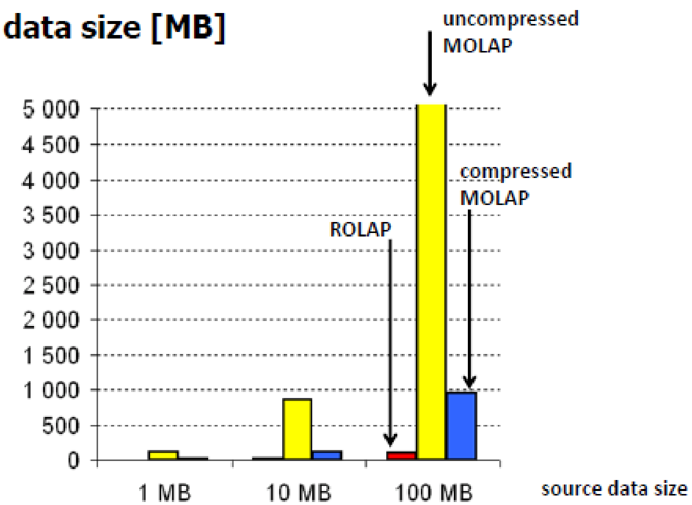
R/M OLAP

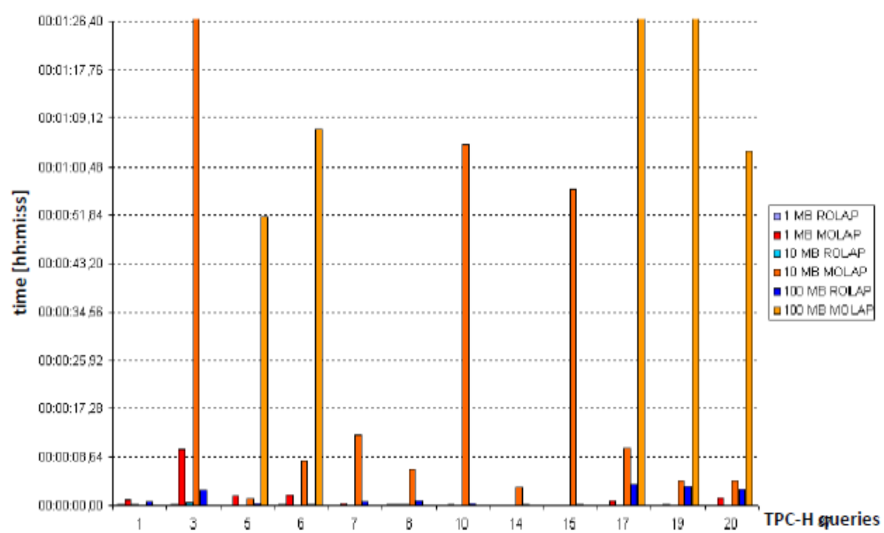
- Oracle
- ROLAP
  - TPC-H benchmark
  - B-tree indexes on PKs and FKs
- MOLAP
  - 4 cubes
  - Discounts(Parts, Orders, ShipTime)
  - Discounts(Parts, Orders, ReceipTime)
  - Quantitites(Parts, Orders, ShipTime)
  - Prices(Parts, Orders, ShipTime)

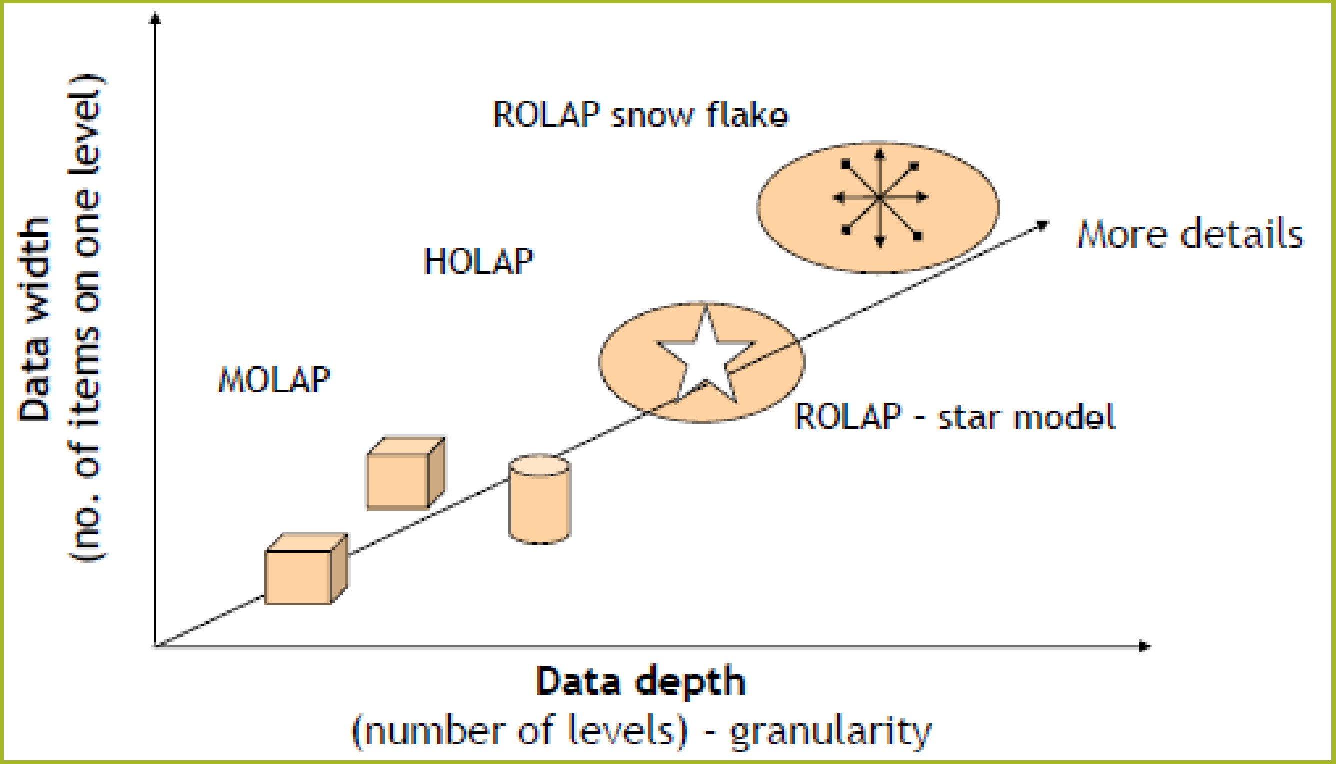
- Load time [hh:mi:ss]



- MOLAP data size [MB]






ROLAP snow flake
HOLAP
MOLAP
More details
ROLAP - star model
Data width
(no. of items on one level)
Data depth
(number of levels) - granularity

R/M OLAP

HOLAP

Presentation layer

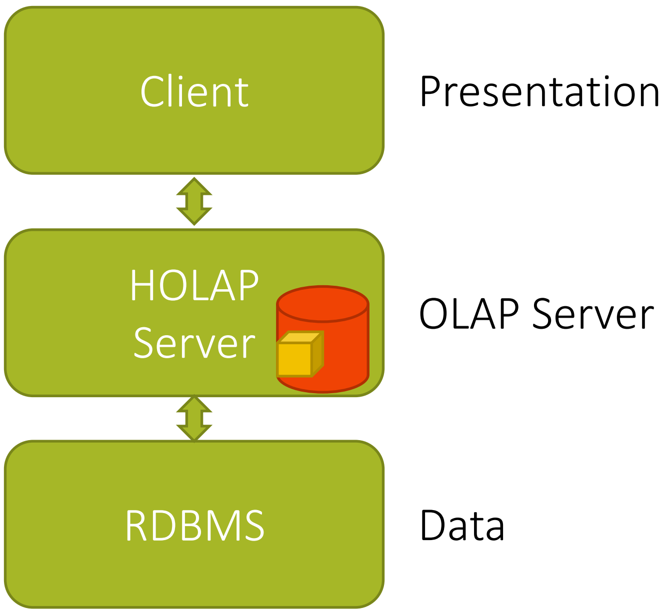
- provides the multi-dimensional view

HOLAP Server

- Storing detailed data in RDBs
- Storing aggregated data in MDBs

RDBMS

- Data is stored in RDBs



HOLAP

- HOLAP model
  - is a tradeoff between MOLAP and ROLAP models
- HOLAP benefits
  - the greater scalability of ROLAP
  - the faster computation of MOLAP
- HOLAP Servers
  - partitioning
    - *horizontal*
    - *vertical*

HOLAP

Vertical partitioning

Aggregations are stored in MOLAP for fast query performance,

Detailed data in ROLAP to optimize time of cube processing (loading the data from the OLTP)

Horizontal partitioning

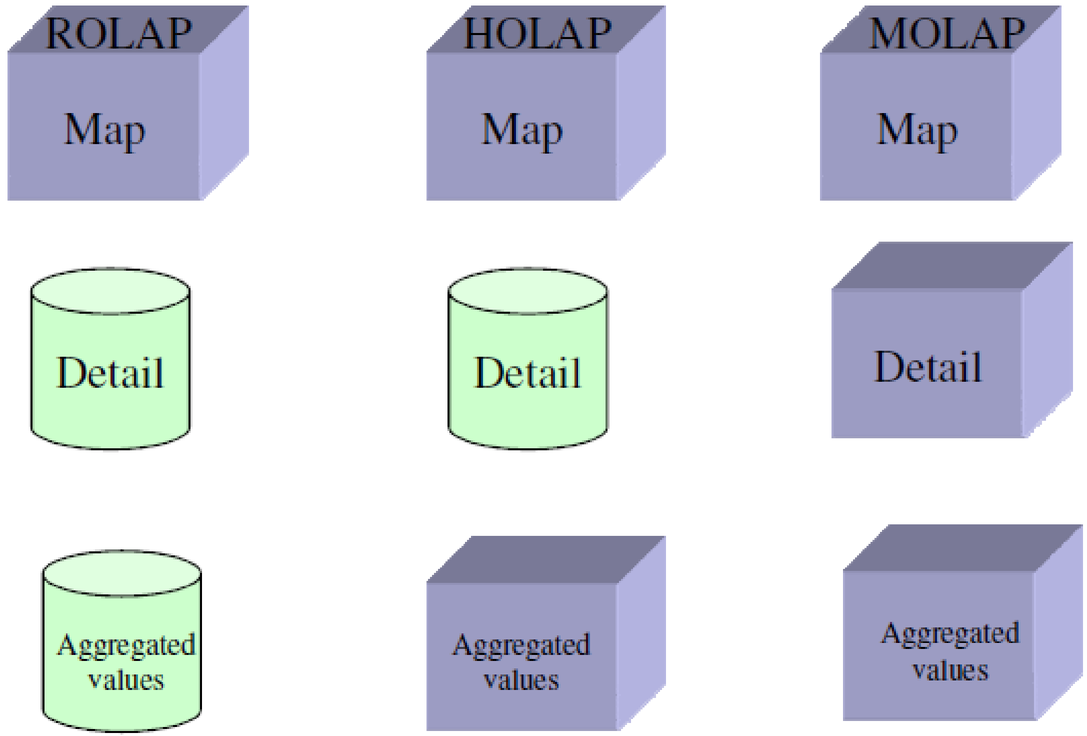
HOLAP stores some slice of data, usually the more recent one (i.e. sliced by Time dimension) in MOLAP for fast query performance

Older data in ROLAP

HOLAP

- MOLAP
  - physically builds “cubes” for direct access
    - *usually in the proprietary format therefore ANSI SQL is not supported.*
- ROLAP
  - provides access to information via a relational database using ANSI standard SQL.
- HOLAP
  - provides a combination of relational database access and “cube” data structures within a single framework.
  - The goal is to get the best of both MOLAP and ROLAP:
    - *scalability (via relational structures)*
    - *high performance (via pre-built cubes)*

OLAP Architecture



R/M/H OLAP

OLAP Server	MOLAP	ROLAP	HOLAP	Offline
Apache Kylin	Yes	No	No	Yes
Druid	Yes	Yes	Yes	Yes
Essbase	Yes	No	No	
IBM Cognos BI	Yes	Yes	Yes	
IBM Cognos TM1	Yes	No	No	Cognos Insight Distributed mode
icCube	Yes	No	No	Offline Cubes
Infor BI OLAP Server	Yes	No	No	Local cubes
Jedox OLAP Server	Yes	No	No	No
Kyligence	Yes	Yes	Yes	No
Kyvos BI on Big Data	Yes	Yes	No	Yes
Microsoft Analysis Services	Yes	Yes	Yes	Local cubes, PowerPivot for Excel, Power BI Desktop
MicroStrategy Intelligence Server	Yes	Yes	Yes	MicroStrategy Office, Dynamic Dashboards
Mondrian OLAP server	No	Yes	No	
Oracle Database OLAP Option	No	Yes	No	
SAP NetWeaver BW	Yes	Yes	No	
SAS OLAP Server	Yes	Yes	Yes	

OLAP server	MOLAP	ROLAP	HOLAP	Offline
Apache Doris	Yes	Yes	Yes	Yes
Apache Druid	Yes	Yes	Yes	Yes
Apache Kylin	Yes	No	No	Yes
Apache Pinot	Yes	Yes	Yes	Yes
ClickHouse	Yes	Yes	Yes	Yes
Essbase	Yes	No	No	
IBM Cognos BI	Yes	Yes	Yes	
IBM Cognos TM1	Yes	No	No	Cognos Insight Distributed mode
icCube	Yes	No	No	Yes
Jedox OLAP Server	Yes	No	No	No
Kyvos	Yes	Yes	Yes	Yes
Microsoft Analysis Services	Yes	Yes	Yes	Local cubes, PowerPivot for Excel, Power BI Desktop
MicroStrategy Intelligence Server	Yes	Yes	Yes	MicroStrategy Office, Dynamic Dashboards
Mondrian OLAP server	No	Yes	No	
Oracle Database OLAP Option	Yes	No	No	
SAP NetWeaver BW	Yes	Yes	No	
SAS OLAP Server	Yes	Yes	Yes	
StarRocks	Yes	Yes	Yes	Yes

https://en.wikipedia.org/wiki/Comparison_of_OLAP_servers

R/M OLAP

A purely relational implementation is designed and optimized to support the efficient movement and calculation of large volumes of data.

- Select a ROLAP implementation when …
  - *The analytic requirements of the business are met by the capabilities of SQL.*
  - *There are appropriate in-house SQL skills.*
  - *The relational engine provides satisfactory query performance*
  - *The detail data is very sparse*
- Select a MOLAP implementation …
  - *When the analytic requirements of the business need the extended analytic, forecasting and planning functionality of Multidimensional Database Technology.*
  - *When the analysis includes lots of calculated and aggregated Key Performance Indicators*
  - *Need an easier way to define complex or proprietary calculations.*

Cubes in SSAS

SSAS

- MOLAP:
  - *Multidimensional OLAP stores a copy of the facts in your SSAS cube.*
    - This is not a one-for-one storage option.
    - Owing to efficient storage mechanisms used for cubes, the resultant storage is approximately 10–20 percent the size of the original data; that is, if you have 1GB in your fact table, plan for around 200MB storage on SSAS.
    - It creates a copy of all source data.
- HOLAP:
  - *Hybrid OLAP does not make a copy of the facts in SSAS.*
    - It reads this information from the star schema source.
- ROLAP:
  - *Relational OLAP does not make a copy of the facts on SSAS.*
    - It reads this information from the star schema source.

SSAS ROLAP Storage

- Aggregations of a partition are stored in indexed views in the relational database specified as the data source of the partition.
  - The indexed views in the data source are accessed to answer queries.
  - If a partition uses the ROLAP storage mode and its source data are stored in SQL Server, Analysis Services tries to create indexed views to store aggregations.
- Query response time and the processing time are generally slower than with the MOLAP or HOLAP storage modes.
  - However, ROLAP enables users to view data in real time and can save storage space when working with large data sets that are infrequently queried, such as purely historical data.

SSAS MOLAP Storage

Both the aggregations and a copy of the source data are stored in a multidimensional structure.

Such structures are highly optimized to maximize query performance.

Query speed

Since a copy of the source data resides in the multidimensional structure, queries can be processed without accessing the source data of the partition.

Note however that data in a MOLAP partition reflect the most recently processed state of a partition.

Thus, when source data are updated, objects in the MOLAP storage must be reprocessed to include the changes and make them available to users.

Changes can be processed from scratch or, if possible, incrementally.

SSAS HOLAP Storage

- Combines features of MOLAP and ROLAP modes:
  - Like MOLAP, the aggregations of the partition are stored in a multidimensional data structure.
  - Like in ROLAP, does not store a copy of the source data.
- If queries
  - only access summary data of a partition
    - *HOLAP works like MOLAP very efficiently.*
  - need to access unaggregated source data
    - *retrieve it from the relational database and therefore will not be as fast as if it were stored in a MOLAP structure.*
      - This can be solved if the query can use cached data (data stored in main memory rather than on disk).
- Partitions stored as HOLAP
  - are smaller than the equivalent MOLAP partitions
    - *since they do not contain source data*
  - can answer faster than ROLAP partitions for queries involving summary data.
    - *Thus, this mode tries to capture the best of both worlds.*

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